

TERM ONE

TOPICS

1. Keeping poultry and bees
2. Measurements
3. Immunisation
4. Digestive system

TOPIC: KEEPING OF POULTRY AND BEES

Poultry keeping

This is the rearing of domestic birds.

POULTRY

These are domestic birds.

Types/examples of poultry

- Chickens
- Ducks
- Turkeys
- Geese
- Guinea fowls
- Pigeons

Examples of poultry products

- Eggs
- Meat
- Feathers
- Poultry droppings
- Bones

Reasons for rearing poultry (Why do farmers keep domestic birds?)

- For egg production/to get eggs.
- For meat production/to get meat.
- For income after sale/to sell them for money.

Importance (uses) of poultry to people

- They provide eggs.
- They provide meat.
- They are sold for money.
- They are used to pay bride price.
- Their droppings are used as manure.
- Their bones are used to make glue.
- Their feathers are used for decoration.
- Their feathers are used to make pillows.
- Their feathers are used to make costumes.

Terms used in poultry keeping

- **Poultry (fowls)** – Domestic birds
- **Hen** – Adult female chicken
- **Cock** – Adult male chicken
- **Cockerel** – Young male chicken
- **Capon** – castrated male chicken

- **Pullet** – young female chicken
- **Chick** – a young bird
- **Incubation** – providing of necessary conditions to a fertilized egg to hatch
- **Incubation period** – time taken by a fertilized egg to hatch
- **Incubator** – a machine used to hatch eggs
- **Layers** – type of chickens kept for egg production
- **Broilers** – type of chickens kept for meat production
- **Dual purpose chickens** – type of chickens kept for both eggs and meat
- **Brooding** – giving of special care to chicks below 8 weeks
- **Broody hen** – a hen incubating eggs to hatch them
- **Brooder** – a special structure where chicks below 8 weeks are cared for
- **Culling** – removal of unproductive birds from the flock
- **Moulting** – Shedding of damaged feathers in birds

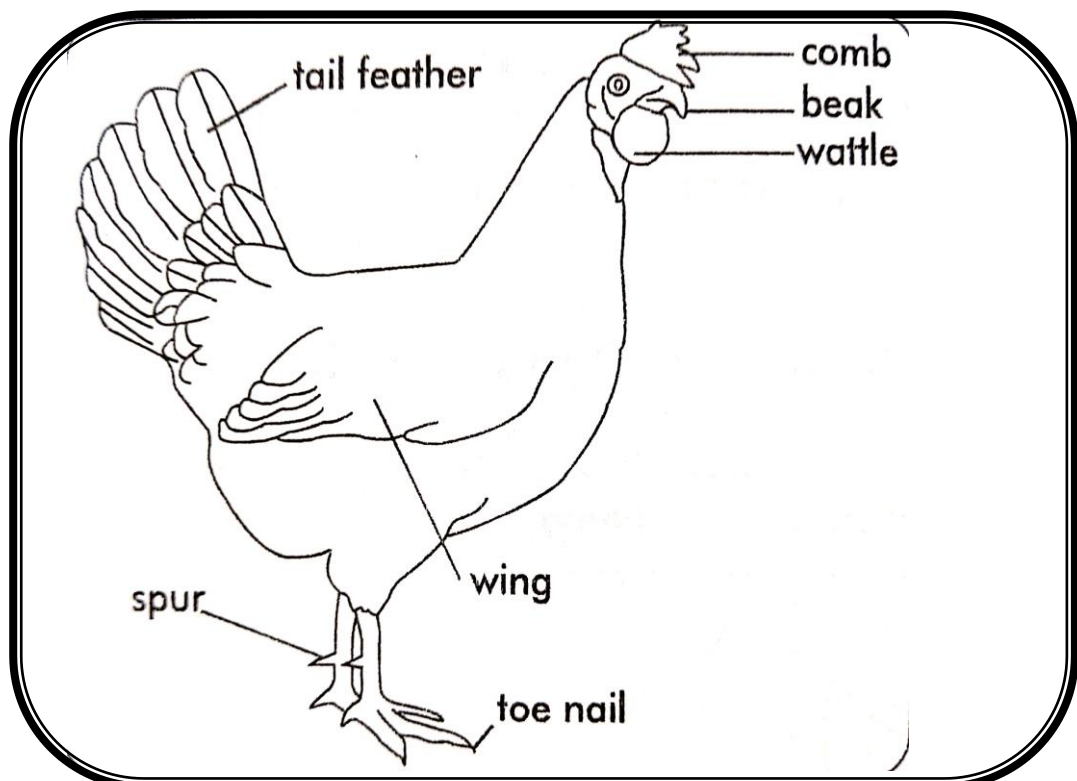
STRUCTURAL DIFFERENCES BETWEEN A HEN AND A COCK

- A hen has a small comb while a cock has a big comb.
- A hen has a small wattle while a cock has a big wattle.
- A hen has a short spur but a cock has a long spur.
- A hen has dull feathers but a cock has bright feathers.

Why do cocks have brightly coloured feathers?

- To attract hens.

EXTERNAL PARTS OF A DOMESTIC BIRD



Functions of each part

Eyes	For sight
Beak	For feeding (for picking up food)
	For protection
Spur	For protection
Nostril	For smelling food
Legs	For walking
Toes with claws	For scratching
Wattle and comb	For temperature regulation

FEATHERS

These are the outermost covers of the bird's body

MOULTING IN BIRDS

- This is shedding of damaged feathers in birds.
- Birds moult once each year.

Why do birds moult their feathers?

- To grow new feathers

Uses of feathers to a bird

- They help a bird to fly (for flight).
- They streamline the bird's body (they give the bird shape).
- They keep the bird's body warm (for warmth).
- They protect the bird's body from injury (for protection against injury).
- They help a male bird to attract mates (for courtship).
- They give the bird colour for identification.

Uses of feathers to people

- They are used as costumes.
- They are used for decoration.
- They are used to make pillows.
- They are used to make mattresses.
- They are used as writing materials.

TYPES OF FEATHERS

- Quill (flight) feathers
- Covert (body) feathers
- Down feathers
- Filoplume feathers

QUILL FEATHERS (FLIGHT FEATHERS)

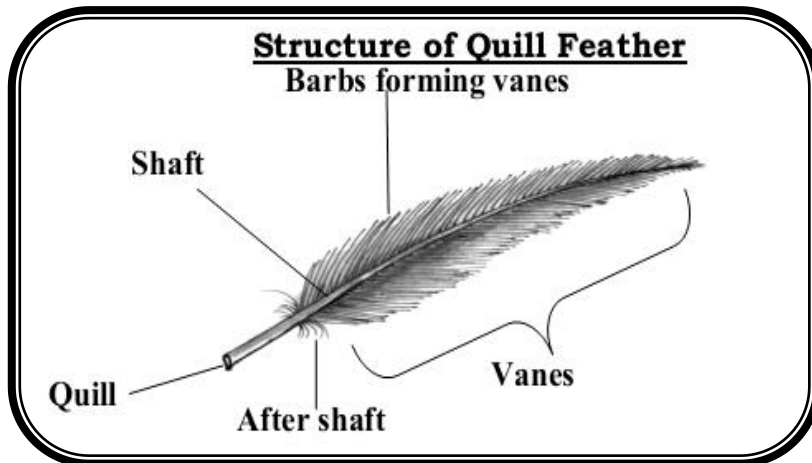
They are found on the **wings** and **tail**.

They are divided into **primary** and **secondary feathers**

Primary feathers are bigger than secondary feathers

Importance of quill feathers

- For flight (they help in flying)



Shaft (rachis)

- It holds the vane (it is where the vane is attached).

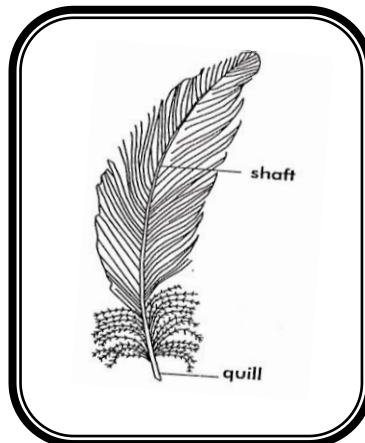
COVERT FEATHERS (BODY FEATHERS)

- They are found on the neck and bases of wings and tail.
- They cover most of the body.
- They are smaller than the quill feathers.

Importance of covert feathers

- They insulate the bird's body.
- They streamline the bird's shape (they give the bird shape).

Structure of a body feather



Why are birds streamlined?

- To overcome viscosity (to reduce air resistance).

What is viscosity (fluid friction)?

- This is the friction in liquids and gases.

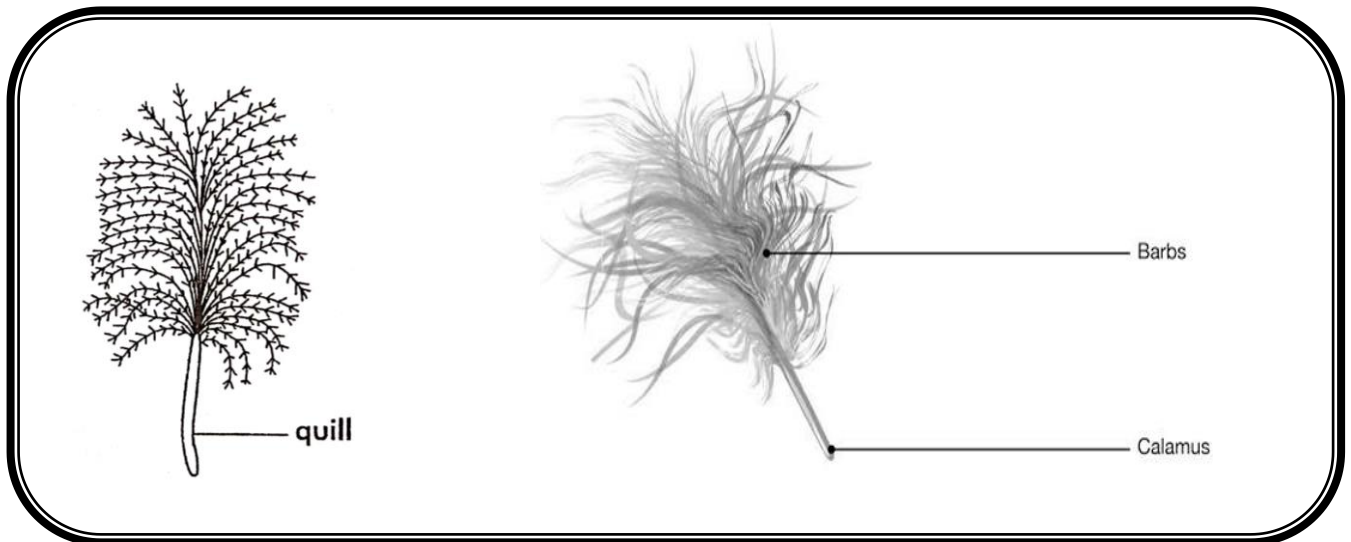
DOWN FEATHERS

- They are found on the **abdominal region**.
- They are the first feathers to appear on a bird.
- They have no vane.
- They have loose barbs.

Importance of down feathers

- They insulate the bird's body.

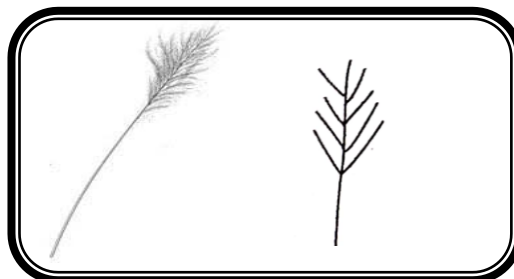
Structure of a down feather



FILOPLUME FEATHERS

- These are the feathers that remain when a bird has been plucked.
- They are found nearest to the skin between the covert feathers.
- They are the tiniest (smallest) feathers.
- They have no quill.

A drawing of a Filoplume feather



Differences between a hen and a cock

Cock

- Has a big comb and wattle
- It has a big and long spur
- Male bird
- Big in size

Hen

- Has a small comb and wattle
- Has a short spur
- Female bird
- Small in size

Types of chicken

- ★ A type of bird is a family of birds kept for specific purpose.
- ★ There are three major types of exotic breeds namely Broilers, layers and Dual purpose birds.

(a) Broilers (heavy breeds)

- ★ These birds are kept mainly for production of meat (chicken). Examples
 - Light Sussex
 - Plymouth Rock
 - Rhode island red
 - New Hampshire
 - Jersey Giant
 - Orpington
 - Sykes

(b) Layers (light breeds)

- ★ Reared mainly for production of eggs, examples
 - White leg horn
 - Brown egger
 - Ancona
 - Minorca

Dual purpose breeds

These are reared for production of meat and eggs.

- Rhode island red
- New Hampshire
- Black Australlorp
- Speckled Sussex
- Brown leghorn
- Turken (Naked Neck)
- Buckeye
- Malines
- Kuroiler

Breeds of chicken

- ★ A breed of chicken is a family of chicken with similar characteristics
- ★ The breeds of chicken include: **A**ncona, **L**ight Sussex, **W**hite leghorn, **R**hode island red, **P**ly mouth rock, **S**ykes, **M**inorca.

Types of breeds of poultry

There are three main or major types of poultry breeds namely: local breeds, exotic breeds, cross breeds.

(a) Local breeds

- ★ They are sometimes referred to as indigenous or native breeds.
- ★ They are called native or local because they existed in Uganda for a very long time.

Characteristics local breeds

- ★ They are resistant to diseases and parasites.
- ★ They have a slower growth rate.
- ★ They lay small and only a few eggs during each laying season.
- ★ They can feed on a variety of feeds.
- ★ They are resistant to bad weather conditions.
- ★ They produce less chicken.
- ★ Have different colours.

Advantages of local breeds over exotic breeds

- ★ Local breeds are more resistant to disease than exotic breeds.

How to improve upon the local breeds of poultry

- ★ By carrying out cross breeding.

(b) Exotic breeds

- ★ These are breeds which were imported into Uganda from other countries e.g white leghorn, Rhode Island red and light Sussex.

Characteristics of exotic breeds

- ★ They are not resistant to parasites and diseases.
- ★ They grow fast and mature in a short time at the same rate.
- ★ They lay many eggs (layers).
- ★ They have the same colour, shape and size.
- ★ Have the same colour.
- ★ They have the same ancestors.
- ★ They produce good quality meat.

Advantages of exotic breeds over local breeds

- ★ They grow and mature faster than local breeds.
- ★ They lay more eggs than local breeds.

- ★ They produce much meat than the local breeds.

Systems of poultry keeping

- Free range system
- Deep litter systems
- Battery cage system
- Pen system/fold system

Free range system

This is when birds are left to move freely looking for their own food.

- ★ In the evening, they come back for shelter.

Why is free range system common in rural areas?

- ★ There is enough land.

Why is free range system not used in urban areas?

- ★ There is inadequate/limited land.

Simple diagram to illustrate a free range system



Advantages of free range system

- ★ It is cheap to manage.
- ★ It saves time.
- ★ Birds get balanced diet.
- ★ Birds get enough body exercise.
- ★ Birds need little care.
- ★ It controls poultry vices.

Why is free range system regarded as the cheapest system of poultry keeping?

- ★ The farmer doesn't buy poultry feeds.

Disadvantages of free-range system

- ★ It needs a big piece of land.
- ★ Birds can easily be stolen.
- ★ Birds can easily be killed by predators/vermins.
- ★ Birds can easily get diseases/there is easy spread of poultry diseases.
- ★ Birds can easily destroy crops.
- ★ Eggs can easily get lost.
- ★ It is difficult to keep farm records.
- ★ It is difficult to cull birds/culling is difficult.

What are vermins?

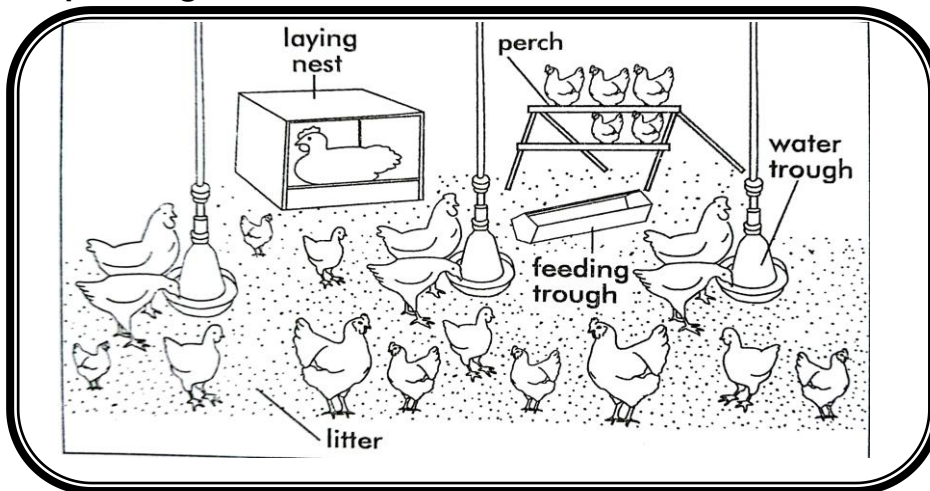
- These are wild animals that attack domestic animals.

Deep litter system

This is when the birds are kept and fed indoors.

- ★ It is an extensive system **because** it needs much attention and labour.

Simple diagram to illustrate



Advantages of deep litter system

- ★ Many birds are kept in a small space.
- ★ Birds are protected from bad weather.
- ★ Birds are protected from predators.
- ★ Birds are protected from thieves.
- ★ Culling is easy.
- ★ It is easy to collect manure.
- ★ It is easy to collect eggs.
- ★ It is easy to keep farm records.

Why is deep litter system regarded as a commercial system of poultry keeping?

- Many birds are kept in a small space.

Why do poultry farmers in urban areas always use deep litter system?

- Due to limited land/there is shortage of land.

Disadvantages of deep litter system

- ★ It is expensive to manage.
- ★ Birds do not get enough body exercises.
- ★ Birds do not get balanced diet.
- ★ It needs much attention.

LITTER

Litter are soft materials put on the floor of a poultry house.

Materials that can be used as litter

- ★ Coffee husks
- ★ Wood shavings.
- ★ Crushed maize cobs

Why is saw dust not recommended to be used as litter?

- It is poisonous when eaten by birds.

Importance of litter

- It prevents dampness of the floor (it keeps the floor of a poultry house warm and dry).
By absorbing moisture from poultry droppings.
- It prevents breaking of eggs.
By acting as a shock absorber.

Qualities of good litter

- ★ It should be clean.
- ★ It should be dry.
- ★ It should not be poisonous to birds.

Battery /cage system

This is when birds are kept in small cages.

- ★ 1 to 3 birds are always kept in each cage.
- ★ The cage has a slightly sloping floor to allow rolling of the eggs into the wire trough.

Advantages of battery cage system

- ★ It prevents contamination of poultry feeds.
- ★ It reduces poultry vices.
- ★ It reduces loss of eggs.
- ★ Eggs laid remain clean.
- ★ Culling is easy.
- ★ Many birds can be kept on a small land.
- ★ Birds are protected from predators/vermins.
- ★ It is easy to collect manure.
- ★ It is easy to keep records of individual birds.

How does battery cage system reduce loss of eggs?

- ★ It prevents contact of birds with their eggs.

Disadvantages of battery cage system

- ★ It is expensive to manage.
- ★ It needs a lot of labour.
- ★ Birds do not get enough body exercises.
- ★ Birds do not get balanced diet.

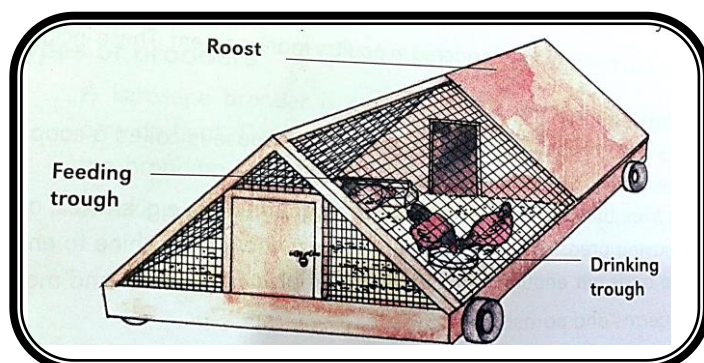
Pen system (fold system)

- ★ This is when the birds are kept inside a small movable house.
- ★ The movable house used in this pen system is called pens, folds or arks.
- ★ The pen or fold is moved to a new place daily.

Importance of a pen to the birds kept inside it

- ★ It provides shade to the birds.
- ★ It protects birds from predators.
- ★ It protects birds from bad weather.

A simple structure of fold, pen system



Advantages of pen system

- ★ Birds cannot destroy crops.
- ★ Birds are protected from predators.
- ★ Birds are protected from bad weather.

- ★ Poultry feeds are not wasted.

Disadvantages of fold pen system

- ★ Few birds are kept.
- ★ Birds do not get enough body exercises.
- ★ Folds get old quickly.
- ★ More labour is needed to move folds every day.
- ★ It is very expensive to make the pen.
- ★ It needs a very big land since folds are moved new places daily.

Poultry management practices (daily/routine activities on a poultry farm)

- | | |
|------------------|--------------------|
| ★ Debeaking | ★ Egg collection |
| ★ Record keeping | ★ Cross breeding |
| ★ Vaccination | ★ Housing |
| ★ Deworming | ★ Feeding |
| ★ Culling | ★ Regular cleaning |
| ★ Dusting | |

DEBEAKING

- This is the shortening of the upper beak of a bird.

An illustration showing Debeaking



Importance of Debeaking

- It controls poultry vices (egg eating, cannibalism and feather pecking).

How does Debeaking control poultry vices?

- It makes the bird's beak blunt.

Danger of Debeaking to a bird

- It can damage the tongue of a bird.

DEWORMING

- This is the giving of drugs to animals to kill worms.

Methods of deworming

- Drenching
- Dosing

Drenching

- This is the giving of liquid drugs to animals through the mouth to kill worms.
- It is done using a **drenching gun**.

Dosing

- This is the giving of solid drugs to animals through the mouth to kill worms.

Importance of deworming

- It controls intestinal parasites in animals (it kills worms in animals).

FEEDING POULTRY

- This is the providing of food to poultry.

Reasons for proper feeding poultry

- To enable birds grow well.
- To enable birds produce more eggs and meat.
- To keep the birds healthy.
- To enable the birds get energy.

Chicken mash

- These are chicken feeds crushed into small pieces.

TYPES OF MASH (CHICKEN FEEDS)

MASH/CHICKEN FEEDS	CHICKENS	AGE
Chick mash/starter mash	chicks	1 day to 8 weeks
Grower mash	Growing layers	8 weeks to 16 weeks
Layers mash	Layers	16 weeks onwards
Broilers mash	Broilers	

Why are poultry feeds always mixed with grit?

- For easy crushing of food in the gizzard.

CHICKEN AND DUCK MASH

- It is fed to a chick from 1 day to 8 weeks.
- A chick needs about 40 grams per day of chick mash.
- Chick mash has a lot of proteins **to give the chick a fast growth**.

GROWERS MASH

- It is fed to growing layers from 8 weeks to about 16 weeks.
- It contains less calcium than the regular layer mash.

LAYERS MASH

- It is fed to layers at 16 weeks onwards.
- It is introduced once the birds have started to lay eggs.

Why are crushed snail shells always added to layer mash?

- To provide calcium to the birds.

Why do layers sometimes lay soft shelled eggs?

- Due to lack of calcium in their diet.

Why should layer mash contain a lot of calcium?

- To enable birds lay hard shelled eggs.

BROILERS MASH

- It is fed to broilers at 8 weeks onwards.
- It enables broilers to produce more meat.

Reasons why birds should be fed on green vegetables?

- To supply birds with vitamins and mineral salts.
- To keep the busy to prevent egg eating.

Why is it not advisable to feed birds on vines and tobacco leaves?

- They are poisonous (toxic) to birds.

Feeding equipment for birds

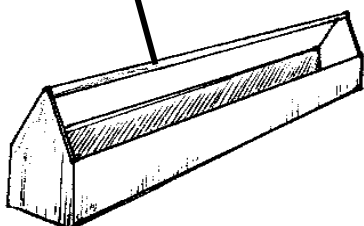
- Feeding trough
It is where chicken feeds are put
- Drinking trough
It is where drinking water for the birds is put

A feeding trough has a spinning/rotating handle **to prevent food contamination by the birds**

Diagrams showing feeding equipment for birds

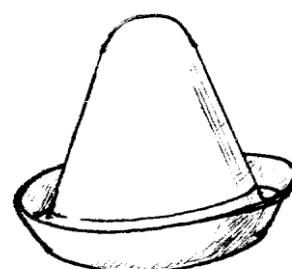
(a) Feeding trough

Spinning/rotating handle



(b) Drinking trough/drinker

conical drinking water trough



Housing poultry

- This is the providing of shelter to birds.

Reasons for proper housing poultry

- ★ To protect birds from bad weather.
- ★ To protect birds from predators/vermins/wild animals.
- ★ To protect the birds from thieves.
- ★ To prevent eggs from getting lost.

Qualities of a good poultry house

- ★ It should be well ventilated.
To allow free air circulation.
- It should have dim light.
To prevent egg eating among layers.
- It should have enough space for the birds.
To prevent poultry vices.
- It should have strong doors.
To prevent attack from predators.
- It should be well roofed.
To prevent leaking of rain water.
- It should be clean and dry.
To prevent easy spread of diseases.

CULLING

- This is the removal of unproductive birds from the flock.

Examples of unproductive birds that should be culled

- Sick birds
- Off layers
- Birds with vices

Ways of culling

- Slaughtering/killing
- Isolation
- Selling

Cross breeding

- ★ This is the mating of a local breed with an exotic breed.
Local breed + Exotic breed = crossbreed/hybrid

Crossbreed/hybrid is the breed got by mating a local breed with an exotic breed.

Why do poultry farmers carry out cross breeding?

- To improve the quality of their breeds.

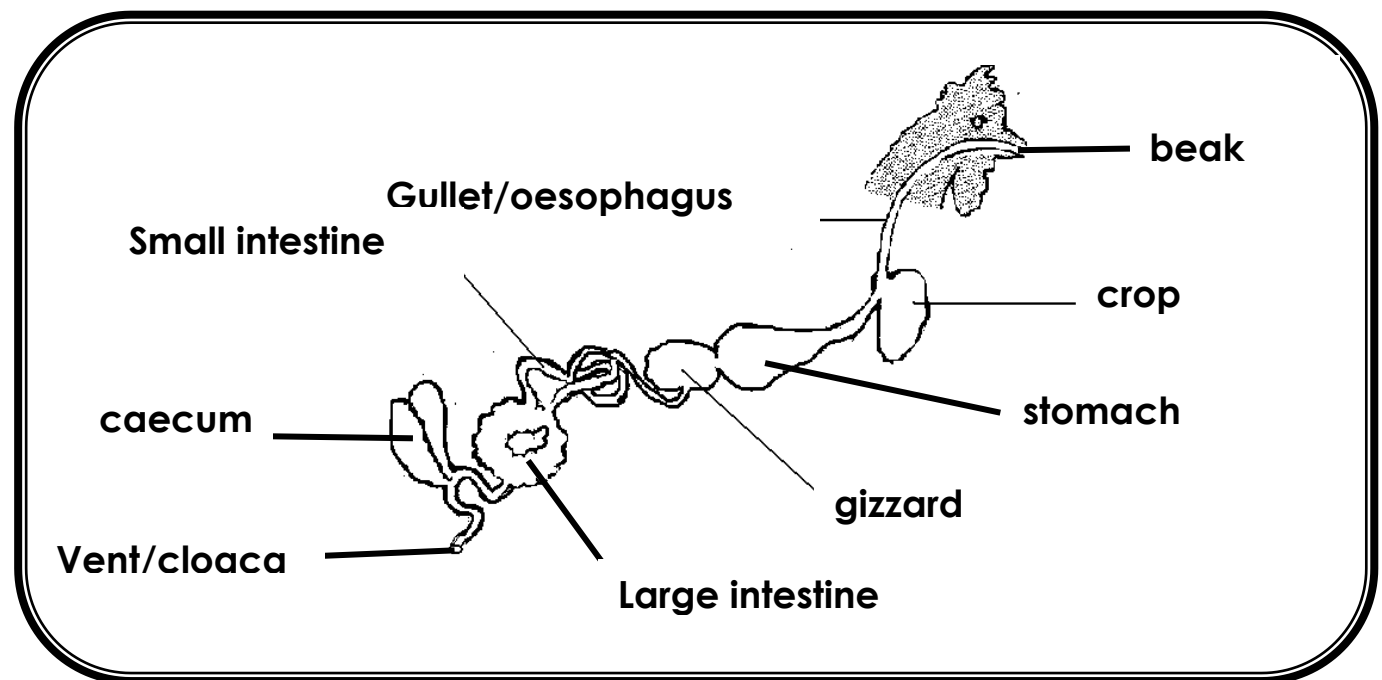
Advantages of crossbreeds over local and exotic breeds

- Crossbreeds grow faster than local breeds..
- Crossbreeds produce more eggs and meat than local breeds.
- Crossbreeds are more resistant to diseases than exotic breeds.
- Crossbreeds are more resistant to bad weather than the exotic breeds.

Ways of improving local poultry breeds

- Cross breeding
- Proper feeding
- Regular vaccination

DIGESTIVE SYSTEM OF A HEN



FUNCTIONS OF EACH PART OF THE DIGESTIVE SYSTEM OF A BIRD

Beak (bill)

- It picks food.

Gullet (oesophagus)

- It passes food to the crop.

Crop

- It stores food for a short time (it stores excess food for later digestion).
- It produces crop milk to feed the chicks e.g in pigeons.

Things that happen to food while in the crop of the bird

- Food is moistened.
- Food is softened.

Examples of birds that do not have a crop

- Owl
- Goose

Stomach (proventriculus)

- It is where food is mixed with digestive juices.
- It secretes digestive enzymes that begin the digestion of proteins.

Gizzard

- It crushes (grinds) food.

How is the gizzard adapted to its function?

- It has grit (small stones).

How is the gizzard able to withstand the grit?

- It has thick (muscular) walls

Which part of the human digestive system perform the same function as the gizzard of a bird?

- Teeth
- Stomach walls

Grit

These are small stones found in the gizzard.

- They crush food into small particles.

Small intestines (ileum)

- It is where food digestion ends.
- It is where food absorption occurs (it absorbs digested food).

Main processes that take place in small intestines

- Food absorption
- Food digestion

Large intestines

- It is where water absorption occurs (it absorbs water).

Caecum

- It stores undigested food for a short time.

Vent (cloaca)

- It passes out droppings.

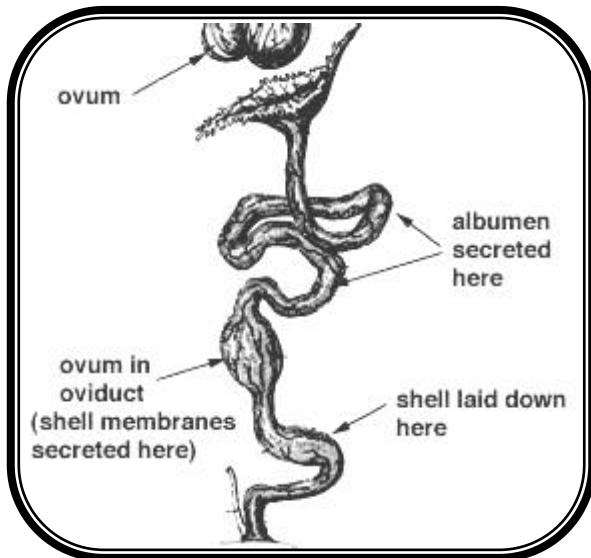
Development of an egg

- ★ An oviduct of a hen consists/contains thousands of OVA (eggs) ovum (one).
- ★ These develop into yolks of the eggs.
- ★ After fertilization the yolk, the white part (egg white or albumen) developed.

Note! Eggs can only be fertilized before the formation of Albumen (egg white).

- ★ The hen under go internal fertilization

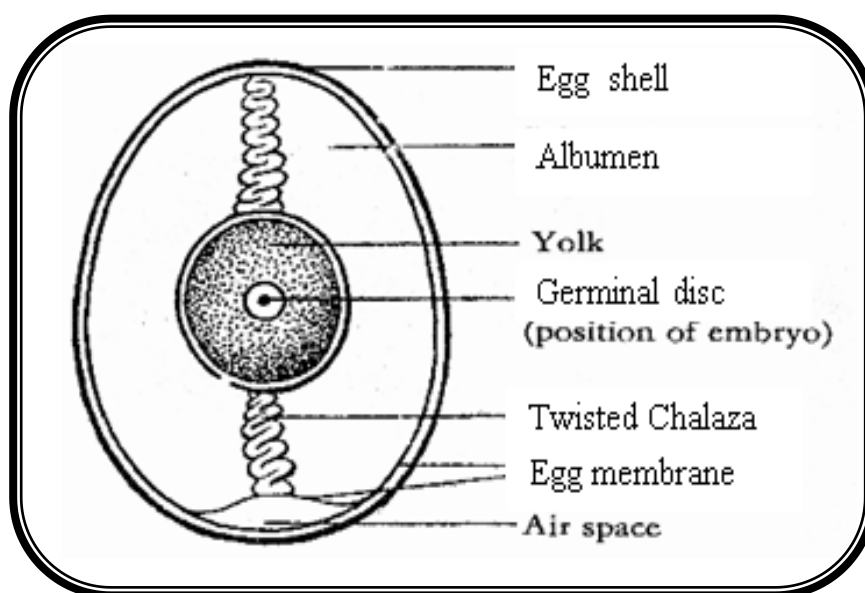
A simple structure of an oviduct of a hen



REPRODUCTION IN BIRDS

- They reproduce by laying eggs (they are oviparous).
- Their eggs are fertilized internally.
- Birds undergo **internal fertilization**.
- Fertilization in birds occurs in the **oviducts**.

STRUCTURE OF A FERTILIZED EGG OF A BIRD



FUNCTIONS OF EACH PART OF A FERTILIZED BIRD'S EGG

Egg shell

- It protects the inner parts of an egg.
- It allows exchange of gases.

How is the egg shell adapted to gaseous exchange?

- It is porous.

How is the egg shell adapted to protection of the inner parts of an egg?

- It is hard.

Why should layers be given mash (feeds) rich in calcium?

- To lay hard shelled eggs.

What enables the porous egg shell to hold the inner components without flowing out?

- ★ It has shell membranes.

Air space

It keeps oxygen for the embryo.

Egg yolk

- It provides fats and proteins to the embryo.

Albumen (egg white)

- It provides water and proteins to the embryo.

Chalaza

- It holds the yolk and embryo in position.
- It is the passage of oxygen to the embryo.
- It is the passage of wastes from the embryo.

Germinal disc

It is found in unfertilized egg.

- It develops into an embryo after fertilization.

Embryo

It is found in a fertilized egg.

- It develops into a young bird.

Functions of an egg to the embryo

- It protects the embryo.
- It provides food to the embryo.

Abnormalities in eggs of a bird

- Blood stained eggs
- Soft shelled eggs
- Double yolked eggs
- Yolkless eggs

Factors that may make a well incubated fertilized eggs fail to hatch

- When the egg is soft shelled.
- When the egg has two yolks.
- When the egg has no yolk.
- When the egg is very dirty.
- When the egg has a crack.

Qualities of a good egg

- It should have an oval shape.
- It should have a porous egg shell.
- It should be hard shelled.
- It must be clean.

INCUBATION

- This is the provision of necessary conditions for a fertilized egg to hatch.

INCUBATION PERIOD

- This is the time taken by a fertilized egg to hatch.

Incubation of different birds

- Pigeons – 16 days (2 weeks, 2 days)
- Hens – 21 days (3 weeks)
- Turkeys – 28 days (4 weeks)
- Ducks – 28 days (4 weeks)
- Geese – 28 days (4 weeks)
- Guinea fowls – 28 days (4 weeks)

INCUBATION

This is the providing of necessary conditions to a fertilized egg to hatch.

Conditions necessary for incubation

- Warmth (optimum temperature)
- Humidity of 60%
To prevent eggs from drying up.
- Good oxygen supply
- Egg turning
To prevent the embryo from sticking to the shell membrane

Types of incubation

- Natural incubation
- Artificial incubation

Natural incubation

★ This is when a hen sits on the eggs to hatch them.

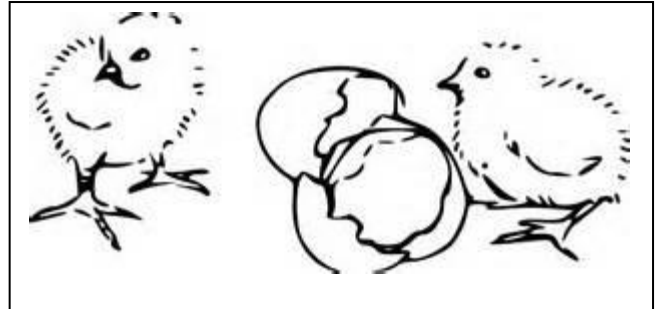
A broody hen is a hen incubating the eggs.

The broody hen usually comes off to feed and gets little exercise.

Diagram of a hen incubating eggs



A chick hatching from an egg



Factors that favour natural incubation

- Use of fertilized eggs.
- Provision of clean dry nest.
- Provision of food to a broody hen.
- Provision of shelter to a broody hen.
- Provision of dim light in the place.
- Protection of the broody from vermins (rats and snakes).

Advantages of natural incubation

- It is cheap and easy to manage.
- Chicks get extra care from the mother hen.
- Chicks get protection from the mother hen.

Disadvantages of natural incubation

- Few chicks are hatched at a time.
- Some eggs are not hatched.
- The mother hen may eat the eggs.
- The mother hen may not be good at incubation.
- Eggs are not tested for their fertility.
- In case the broody hen dies, the eggs may not hatch.

Why is natural incubation not good for commercial farmers?

- Few chicks are hatched at a time

Artificial incubation

This is when an incubator is used to hatch eggs.

- **An incubator** is a machine used to hatch eggs.

Types of incubators

- Electric incubators
- Kerosene incubators

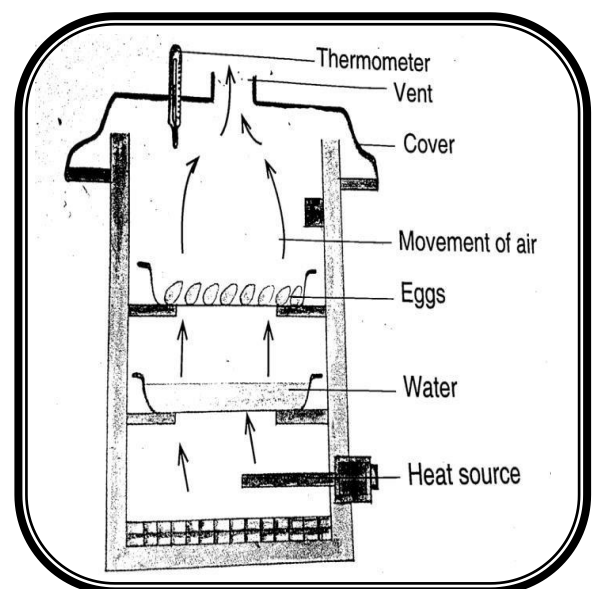
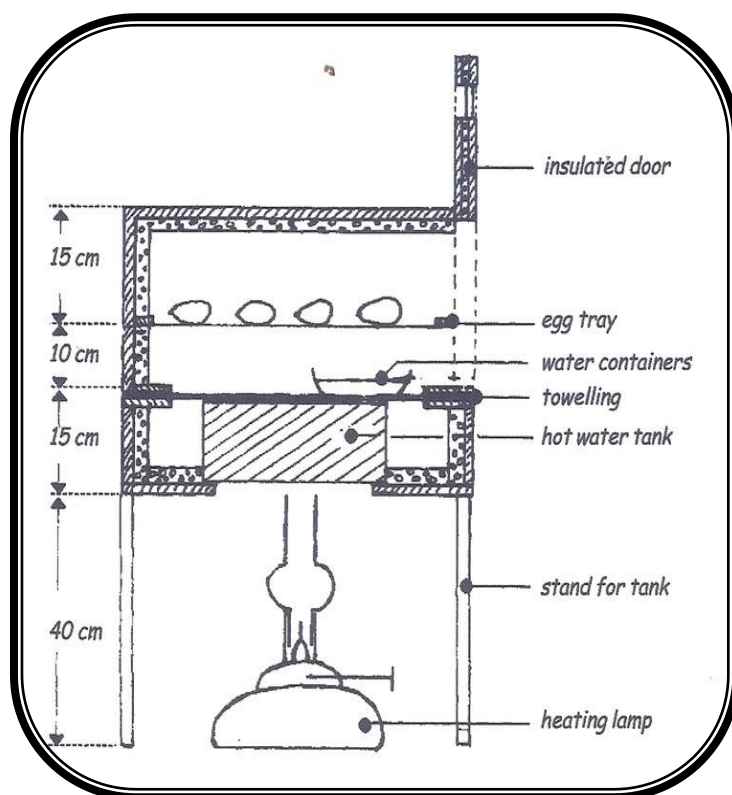
Electric incubators

- These are incubators that use electricity to provide warmth

Kerosene incubators

- These are incubators that use kerosene to provide warmth

An incubator



A clutch:

- This is a group of eggs in a nest or incubator.

Advantages of Artificial incubation

- Very many eggs are hatched at a time.
- Eggs are tested for their fertility.
- All eggs are hatched.

Why is artificial incubation good for commercial farmers?

- Very many eggs are hatched at once

Disadvantages of artificial incubation

- It is expensive.
- It needs much attention.
- It needs skilled labour.
- It is tiring.
- Eggs can be affected in case of frequent black out or load shedding.

BROODING

- This is the giving special care to chicks below 8 weeks.

Examples of special care given to chicks

- Food
- Warmth
- Shelter

Types of brooding

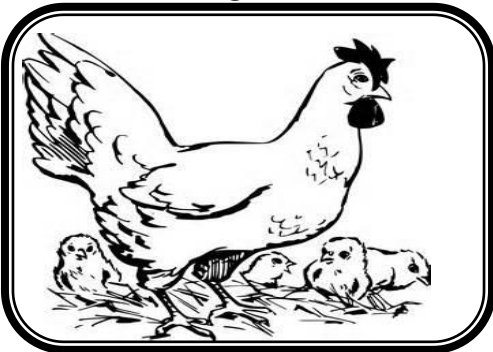
- Natural brooding
- Artificial brooding

Natural brooding

This is when a hen takes more care of her chicks.

- ★ Farmers may also provide more food and security to the mother hen and chicks.

A hen looking after her chicks



Advantages of natural brooding

- It is cheap.
- It reduces poultry vices among chicks.
- The chicks get security from the mother hen.

Disadvantages of natural brooding

- Chicks can be killed by bad weather.
- Chicks can be eaten by predators.
- Few chicks are cared for.
- It gives little profits.
- In case the hen dies, chicks are left without care.

Artificial brooding

This is when chicks are kept in a brooder.

★ A brooder is a special structure in which chicks are cared for.

Types of brooders

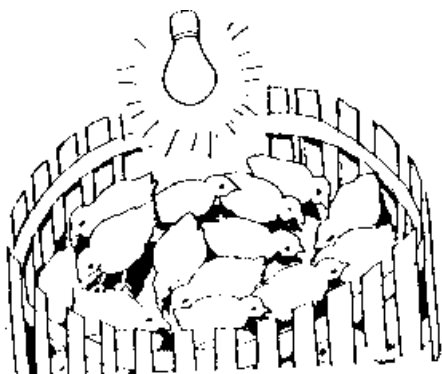
- Infrared brooder
- Kerosene brooder
- Charcoal brooder

Infrared brooder

- Infrared brooder

In this types electricity is used as the source of heat.

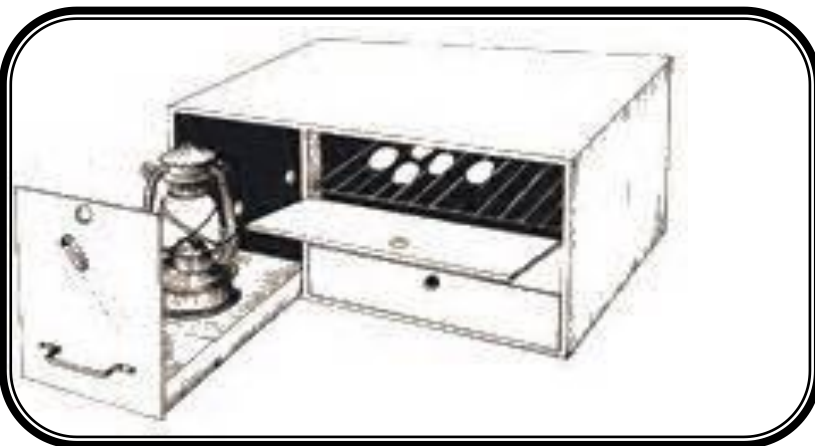
A simple diagram of infra-red brooder



Importance of infrared lamp

- ★ It provides light
- ★ It provides warmth

A simple diagram of Kerosene brooder



- The lamp has guards to prevent chicks from being burnt

Dangers of a kerosene lamp

- It may burn the chicks.
- It leads to accumulation of soot.

Poultry vices (fowl vices)

- These are bad habits in poultry.

Examples of poultry vices

- Feather pecking
- Toe pecking
- Cannibalism
- Egg eating

Cannibalism

- This is when a bird eats flesh of other bird.

Egg eating

- This is when layers eat eggs.

Feather pecking

- This is when a bird uses a beak to pull off feathers of other bird.

Causes of poultry vices

- Overcrowding of birds.
- Boredom among birds.
- Too much light around laying nests.
- Lack of poultry feeds.
- Failure to collect eggs in time.
- Failure to provide enough poultry feeds.
- Failure to remove broken eggs from poultry house.

Control of poultry vices

- By Debeaking.
- By culling.
- By providing enough space for the birds.
- By providing enough food to poultry.
- By collecting eggs in time.
- By providing dim light around laying nests.
- By removing broken eggs as soon as possible.
- By providing green vegetables to birds.

Poultry diseases

- These are diseases that affect domestic birds

Causative agents of poultry diseases (types of germs that cause poultry diseases)

- Virus
- Bacteria
- Protozoa

EXAMPLES OF POULTRY DISEASES

Viral poultry diseases

- ★ Fowl pox
- ★ Avian leucosis
- ★ Newcastle disease
- ★ Gumboro
- ★ Marek's disease

Protozoan poultry diseases

- Coccidiosis
- Blackhead

Bacterial poultry diseases

- Fowl typhoid
- Pneumonia

Causes of poultry diseases

- Poor sanitation in the poultry house.
- Poor feeding of poultry.
- Poor housing of poultry.
- Poor ventilation of the poultry house.

Control of poultry diseases

- Regular vaccination.
- Regular cleaning of the poultry house.
- Culling
- Proper ventilation of the poultry house.
- Proper housing of poultry.

Effects of diseases in poultry

- Death of poultry
- Poor growth of poultry
- Drop in egg production
- Poor quality of eggs and meat

Parasites of poultry

(a) What is a disease?

It is simply an illness in human beings, animals and plants that can be caused by a germ.

(b) **What is a parasite?**

This is a living organism which depend on a host for food and shelter.

General causes of diseases and parasites

- ★ Poor feeding of poultry on already contaminated food.
- ★ Failure to carryout important farm operations on time such as vaccination, Deworming, culling, Dusting.
- ★ Poor housing structure for birds.
- ★ Dirty houses/structure may lead to un out break of diseases and parasites.
- ★ Lack of technical knowledge (skilled man power).

Dangers of parasites

- ★ Death may occur.
- ★ Lower the quality of products.
- ★ Production of meat (chicken) and eggs is greatly reduced.
- ★ Reduces the bird's rate of growth.

Parasites

These are living organism which depend on a host for food and shelter.

- ★ This may happen in both plants and animals
- ★ There are two types of animal parasites namely

(a) Ectoparasites (External parasites)

(b) Endo – Parasites (internal parasites)

Ectoparasites

These are parasites which attack the animals from the outside of the body e.g lice, red mites, fleas Bed bugs, Ticks

Endoparasites

- ★ These are parasites which attack the internal body organs.
- ★ For case of poultry, egg production is greatly affected thus leading to disease attack.
- ★ Common examples are: Round worm, Hookworm, Tape worm, Ring worm, Threadworm.

Types of parasites	Examples	Effects
Ecto parasite	Fleas Chicken mite Fowl tick	★ They create wounds on skin of birds ★ They cause itching and scratching of body that may lead to loss of feathers ★ They suck blood from the birds making them anaemic ★ Egg laying is greatly reduced ★ Mites cause a disease called Mange
Endo parasites	Tape worm Round worm	★ They cause stunted or poor growth ★ Egg laying is reduced

General control/Treatment of parasites in poultry

- ★ Use of pesticides for external parasites.
- ★ Carry out regular deworming to kill internal parasites (dewormers).
- ★ Clean poultry houses regularly dirty houses may lead to accumulation of diseases.
- ★ Dip the feet in paraffin to control scale.
- ★ Apply Vaseline jelly on affected parts of the body caused by mites and fleas.
- ★ Turn the litter regularly to dry in case of wet litter (wet litter can lead to accumulation of parasites).

FARM RECORDS

- These are written information showing different activities done on a farm

Types of farm records

- ★ Breeding records
- ★ Production records
- ★ Health records
- ★ Labour records
- ★ Sales and expenses records

- ★ Inventory records
- ★ Feeding records

Reasons for keeping farm records

- ★ To know the profits or losses.
- ★ To be taxed fairly.
- ★ To get loans easily.
- ★ To plan for the farm.
- ★ To know the farm history.
- ★ To make proper decision.

SOCIAL INSECTS

These are insects that live and work together.

Examples of social insects

- ★ Honeybees, bumble bees
- ★ Wasps
- ★ Termites
- ★ White ants

GROUPS OF BEES

- ★ Solitary bees
- ★ Social bees

Solitary bees

This is a group of bees that live and work alone.

Examples of solitary bees

- Mason bees
- Mining bees
- Sweat bees
- Leafcutter bees
- Carpenter bees

SOCIAL BEES

- These are bees that live and work together.
- A group of bees is called a **bee colony**.
- A group of bees in motion is called a **bee swarm**.

Examples of social bees

- Honeybees
- Bumblebe

APICULTURE

- This is the keeping of honey bees.

Importance of bees to plants and people

People

- ★ Bees provide honey.
- ★ Bees provide beeswax.
- ★ Bees provide pollen and propolis.
- ★ Apiculture project is a source of employment.

Plants

- Bees pollinate flowers of crops

Danger of bees in the environment

- They sting people and animals.

What causes bees to sting?

- ★ Honey harvesting during day time.
- ★ Honey harvesting without a smoker.
- ★ Using cosmetics made from beeswax.
- ★ Making a lot of noise near the hive.
- ★ Standing in path of bee swarm.
- ★ Playing near the hive.
- ★ Crushing a bee near a hive.
- ★ Chasing away bees with naked hands.

MAIN BEE PRODUCTS (MAIN PRODUCTS GOT FROM HONEY BEES)

- ★ Honey
- ★ Beeswax

Other bee products

- ★ Propolis: it is used to heal cuts and wounds
- ★ Bee pollen: it is rich in proteins
- ★ Bee venom
- ★ Royal jelly: it is rich in vitamin B

Uses of honey to people

- ★ It is used as food.
- ★ It is used to sweeten tea.
- ★ It is used as medicine for burns and cough.
- ★ It is used to make alcohol.
- ★ It is used in baking (to make breads).
- ★ It is a source of income when sold.

Why does honey take long to get spoilt?

- ★ It is acidic and contains little moisture.

Which food values do we get from the following bee products?

- ★ **Honey:** Carbohydrates
- ★ **Pollen:** Proteins
- ★ **Royal jelly:** Vitamins

BEESWAX

- It is produced by wax glands inside the body of a honey bee.
- ★ Beeswax is used by bees to build honey combs.

Products from beeswax

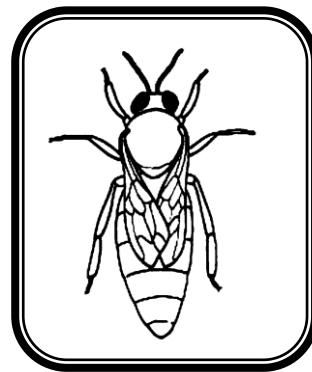
- | | | |
|---------------|---------------|-------------|
| ★ Wax candles | ★ Skin cream | ★ Cosmetics |
| ★ Shoe polish | ★ Chewing gum | ★ Crayons |
| ★ Lip balm | ★ Aftershave | |

TYPES (CASTES) OF BEES IN A BEE HIVE

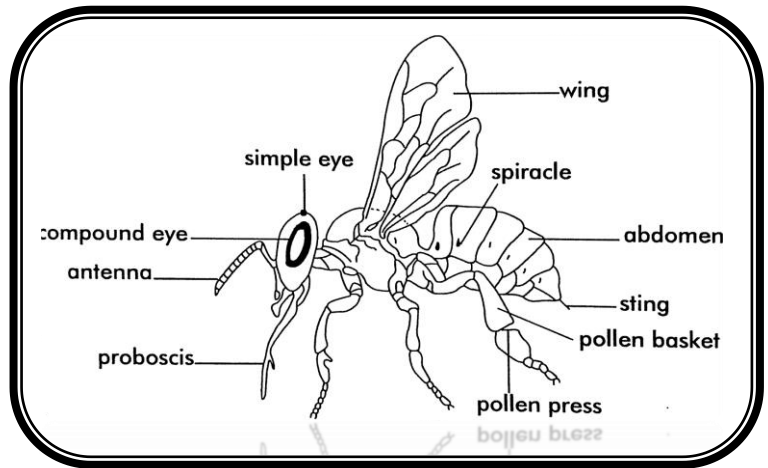
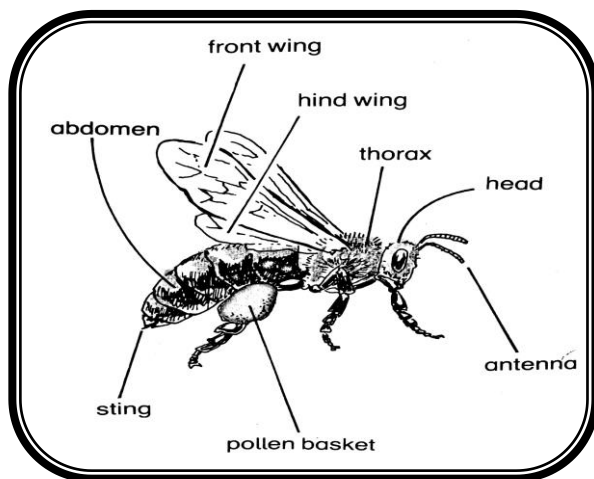
- ★ Worker bees
- ★ Drone bees
- ★ Queen bees

Queen bee

- ★ It is the mother bee in the bee hive .
- ★ It is head of the colony.
- ★ It develops from a fertilised egg.
- ★ Its main function is **to lay eggs** (She lays up to 1500 eggs a day).
- ★ It is a fertile female bee **because** it has an ovipositor for laying eggs.
- ★ It is the largest bee in the hive.
- ★ Her abdomen and legs are longer than others.
- ★ It has an **ovipositor** for laying eggs.
- ★ Its fed on special food called **royal jelly** by worker bees.
- ★ Royal jelly is produced by the salivary glands of worker bees.
- ★ The queen bee mate once in her life time and the sperms are stored in the sperm sac in her abdomen.
- ★ The flight when a drone bee mates with the queen bee is called **mating flight (nuptial flight)**
- ★ There is only one queen bee in each hive.
- ★ As soon as a new queen is hatched, the old one leaves the hive (swarming occurs).



Simple diagram to illustrate



Worker bees

- ★ Worker bees are sterile female bees **because** they lack an ovipositor.
- ★ They develop from fertile eggs.
- ★ They are the smallest bees in size.
- ★ They have a stinger **for protection**.
- ★ Worker bees die after stinging **because** its abdomen is damaged when the stinger is lost.
- ★ Worker bees make up the biggest number in the hive **because** they perform all the duties in the hive.
- ★ They have a pollen basket **for carrying pollen (and propolis)**.

Roles of a worker bees in the hive

- ★ They feed the queen, drone and grubs.
- ★ Worker bees which feed the queen are called **Nurse bees**.
- ★ They guard the hive.
- ★ They collect nectar, pollen, propolis and water.
- ★ They make honey.

Honey is made from water, nectar and pollen

- ★ They clean the hive.
- ★ They repair the hive.
- ★ They build honey combs and brood combs.
They use wax to build honey combs
- ★ They regulate temperature inside the hive.
By fanning the hive
- ★ They look for a new site.

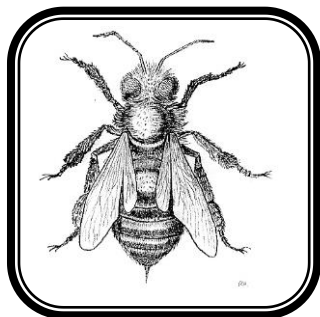
PROPOLIS

This is a sticky substance collected by worker bees from plants.

Uses of propolis to bees

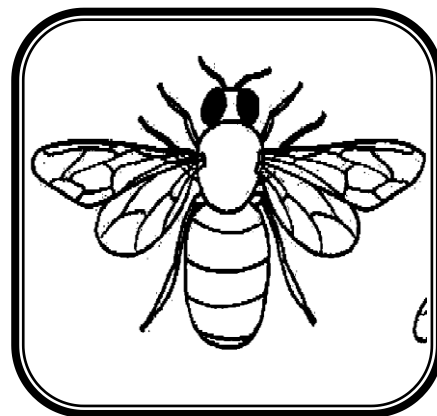
- ★ For repairing the hive.
- ★ For smoothing the hive.
- ★ For trapping dust and germs.
- ★ For making the hive water proof.

Simple diagram of a worker bee



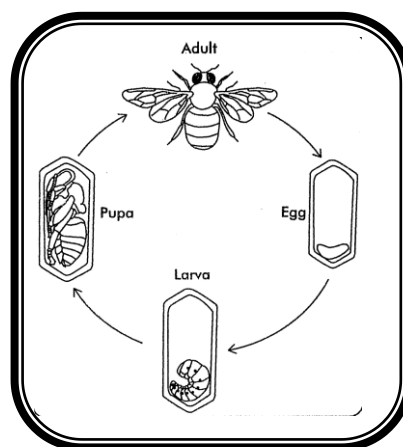
Drone bee

- ★ It is a male bee in the hive.
- ★ It develops from **unfertilized eggs**.
- ★ It has a stout shape.
- ★ It is the second biggest in the hive.
- ★ Its main work is **to mate with the queen bee**.
- ★ It has a broad blunt (round bottomed) abdomen.
- ★ It has no stinger.
- ★ It makes a buzzing sound when flying.
- ★ It dies after mating **because** its male organs break off during mating hence damaging the abdomen.
- ★ Drones are rarely found in the hive **because** they are stung to death by worker bees during food shortage.



The life cycle of a honey bee

- ★ Honey bees undergo **complete life cycle** (4 stages of development)
- ★ The queen bee lays eggs.
- ★ Eggs develop into larvae.
- ★ Bee larvae are called **grubs**.
- ★ **Grubs** are fed by worker bees.
- ★ Larvae develop into pupae.
- ★ Pupae develop into adult bees.



Types of eggs laid by queen bee

- Fertilized eggs
- Unfertilized eggs

Fertilized eggs develop into **queens** and **worker bees**.

Unfertilized eggs develop into **drone bees**.

The adult stage in the life cycle of insects is called **imago**.

What makes fertile eggs to either develop into a queen or worker bee?

- ★ The change of food during larvae stage.
- ★ Larvae to develop into worker bees are fed on **honey** and **pollen**.
- ★ Larvae to develop into queen bees are fed on **royal jelly**.
- ★ Larvae and pupae are found in **brood combs**.
- ★ Honey is kept in **honey combs**.
- ★ Bees use wax to make combs.
- ★ The pupae develop into **adult bees**.

SWARMING

- This is the massive movement of bees from one place to another looking for a new hive.

What is absconding in bees?

- This is condition when all bees move from the hive looking for a new hive.

Reasons for swarming in bees

- | | |
|--------------------------------------|--|
| ★ When the hive is damaged (leaking) | ★ Direct smoke into the hive. |
| ★ When bees are attacked by enemies | ★ Direct sunshine into the hive. |
| ★ When a new queen bee develops. | ★ Lack of flowering plants and water in an area. |
| ★ Bad smell around the hive. | ★ Dampness of the bee hive. |
| ★ A lot of noise around the hive. | |
| ★ Overcrowding of bees in the hive. | |

How to prevent swarming

- | | |
|---|--|
| ★ Siting a beehive near flowering plants. | ★ Siting a beehive in a quiet place. |
| ★ Siting a beehive near open water sources. | ★ Protecting the beehive from direct sunshine. |
| ★ Feeding bees during drought conditions. | ★ Protecting bees from their enemies. |

BEE HIVE

- ★ This is a habitat for bees.
- ★ This is a structure in which bees are kept.

Types of beehives

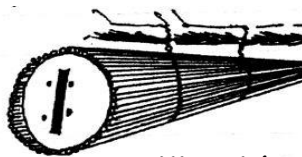
- Traditional (local) beehive
- Modern beehive

(a) Traditional bee -hive

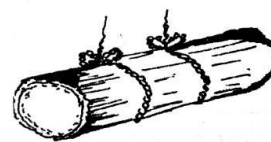
These are bee hives made out of local materials such as hollow logs, grass, woven sticks, cow dung and reeds.

Examples of local hive

- ★ Kigezi beehive
- ★ Dug out log hive
- ★ Tin hive



Kigezi beehive



Dug out log hive

Advantages of traditional (local) bee hive

- ★ It is cheap.
- ★ The colony is not always disturbed by a bee keeper.
- ★ It is easy to make.

Why?

- Materials are locally available

Disadvantages of local bee hive

- ★ Honey is always mixed with eggs.
- ★ The hive is damaged during honey harvesting.
- ★ It is difficult to inspect honey combs.
- ★ It is difficult to prevent swarming.
- ★ The brood is disturbed during honey harvesting.
- ★ It is wasteful.

(b) Modern bee hives

These are hives made from modern materials e.g tins, wire mesh and iron sheets.

Examples of modern bee hives

- ★ Top bar hive
- ★ Box hive

Modern bee hives have two chambers

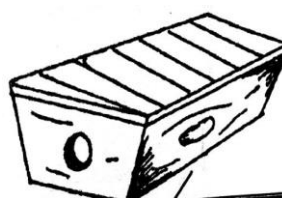
- ★ Brood chamber
- ★ Honey chamber

Queen excluder

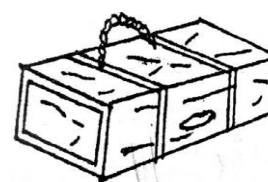
- ★ This is the structure that separates the brood chamber and the honey chamber.
- ★ It has small holes for only worker bees to pass through and look after the queen
- ★ The queen and brood are found in the **brood chamber**.
- ★ The queen cannot move to the honey chamber **because** it is too big to pass through the small holes in the queen excluder.
- ★ Honey is kept in the honey chamber.

Importance of queen excluder

- ★ It prevents the queen from laying eggs in honey

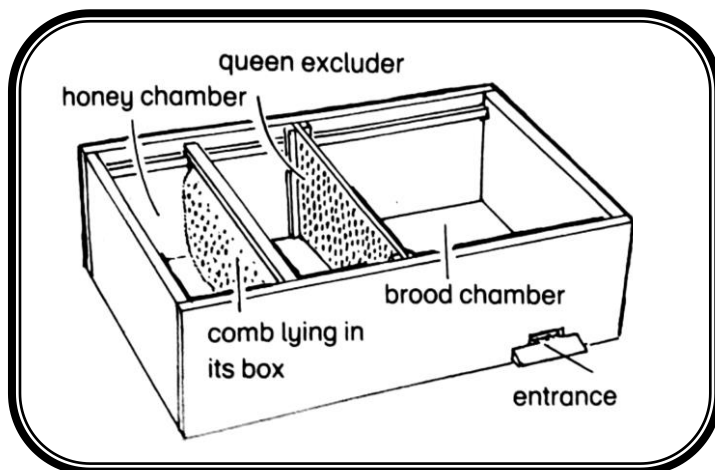


Top bar hive



Box hive

Internal structure of a modern bee-hive



Advantages of modern bee hives

- ★ Clean honey is harvested.
- ★ Modern hives are durable (long lasting).
- ★ The hive is not damaged during honey harvesting.
- ★ It is easy to inspect honey combs.

Disadvantages of modern bee-hive

- It is expensive to make.

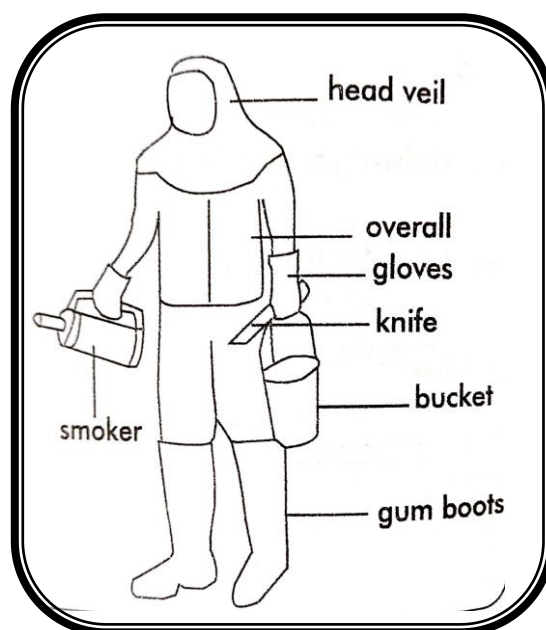
HARVESTING HONEY

- ★ This is the removal of honey combs from the bee hive
- ★ Honey must be or should be harvested in the evening when it is cool and all the bees have settled inside the hive
- ★ When harvesting honey, the farmer must have the following equipment

- (a) A bee veil
- (b) A bee keeper's glove
- (c) A bucket
- (d) A suitable protective clothing (Overall)
- (e) A smoker
- (f) Gum boots
- (g) Knife

Uses of each equipment above

- **Bucket:** it is used for collecting honey.
- **Bee veil:** protects the face of a honey harvester from being stung by worker bees.



- **Bee keeper's gloves:** protect hands of a honey harvester from being stung by worker bees.
- **Smoker:** for smoking the hive to calm the bees.
- **Knife:** for cutting honey combs.
- **Gum boots:** To prevent worker bees from stinging the feet of a honey harvester.

Steps followed when harvesting honey

1. Puff smoke into the hive To tame the bees
2. Lower the hive To prevent damaging the honey combs
3. Open the hive
4. Cut the honey combs

Some honey combs should be left in the hive **for the worker bees to feed on as they build new ones**

Disadvantages of harvesting honey at night

- The brood may be crushed at contaminate honey.
- It is difficult to distinguish between brood combs and honeycombs.

Factors to consider when harvesting honey

- ★ Don't frighten the bees while harvesting.
- ★ Don't kill bees while harvesting.
- ★ Dress properly in suitable clothes.
- ★ Use a smoker.
- ★ Bees should not be crushed while harvesting honey.
- ★ Only remove combs capped with honey.
- ★ Don't take any honey comb containing brood.

Honey extraction

- This is the removing of honey from honey combs.

Method of extracting honey

Methods:

1. Centrifuging method.

A centrifuging machine is used to remove honey from the combs.



Centrifugal honey extractor

2. Pressing the honey method.

Honey combs are put in a cloth sieve and then pressed to remove honey.

3. Floating the wax method.

Honey combs are placed in a clean saucepan which is later put in a bigger saucepan containing hot water. The heat from the water makes honey in a clean container to melt thus making the wax to float.

Strainer is a machine used to remove impurities from honey

CARE AND MANAGEMENT OF BEES

SITING THE BEE HIVE

- This is the selecting of a suitable place where to put a bee hive.

APIARY

- This is a place where many bee hives are kept or set.

Factors to consider when siting hives

- ★ Bee hives should be kept in quiet places (bees don't like noisy places).
- ★ Bee hives should be protected from direct sunshine and wind.
- ★ Bee hives should be in places near water source.
- ★ Bee hives should be in places near flowering plants.
- ★ Bee hives should not be near a school.
- ★ Bee hives should not be near farm animals.
- ★ Bee hives should not be near market place.
- ★ Bee hives should not be near main road and houses.
- ★ Bee hives should be in places with short grass.

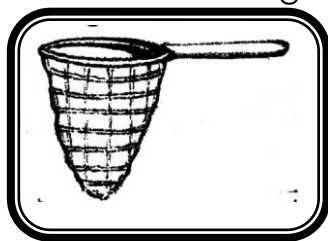
STOCKING THE HIVE

- This is the act of putting bees in an empty hive
- This is the act of encouraging bees to occupy an empty hive

Ways of stocking the hive

- ★ Use of a catcher box
- ★ Use of a swarm catching net
- ★ Smearing the inside parts with wax and wait bees to come for their own.

NB: Its handle can be made longer to catch a higher swarm



BAITS IN BEE KEEPING

- These are things used to attract a swarm of bees into a new hive

Examples of baits in bee keeping

- ★ Beeswax
- ★ Syrup (sugar solution)
- ★ Lime juice
- ★ Granulated sugar

FEEDING BEES

- Bees naturally feed on **nectar** from flowers

Conditions that make bees be fed on supplementary food

- When they are new in a hive
- During the drought conditions
- To encourage multiplication

Examples of supplementary feeds

- Syrup (ordinary sugar mixed with water)
- Ordinary water put in a container at a distance of 10m from the hive

HANDLING OF BEES BY A BEE KEEPER

(Three occasions when a bee farmer handles bees)

- When harvesting honey.
- When inspecting honey combs.
- When stocking the hive.

ENEMIES OF BEES (PESTS)

(a) Safari ants and red ants

How to protect bees from ants

- ★ Hang the hive between poles.
- ★ Smear oil at the bases of poles.
- ★ Put grease on the wires, This can prevent the ants from reaching the hive.

(b) Honey Badgers

- These spoil the hives, kill bees and eat the honey

OTHER BEE PESTS (ENEMIES OF BEES)

- ★ Wax beetles
- ★ Wax moths
- ★ Wasps
- ★ Mockingbirds

DISEASES OF HONEY BEES

- Varroosis
- American foulbrood
- Chalkbrood
- Sacbrood

Control of bee pests and diseases

- ★ Keep the hive free from dampness.
- ★ Disinfect a hive where bees have died.

- ★ Burn all infected combs.
- ★ Repair broken hives to prevent honey badgers.

TOPIC TWO: MEASURES

Regular shaped objects

- These are objects with defined shape (definite shape)

Examples of regular objects

- ★ Cube
- ★ Cuboid (brick, box, block)
- ★ Cylinders
- ★ Cone

Measuring regular shaped objects.

(a) Length

This is the distance between points

Units for measuring length

- ★ millimetres (mm)
- ★ centimetres (cm)
- ★ decimetres (dm)
- ★ metres
- ★ decametres (Dm)
- ★ Hectometres (Hm)
- ★ Kilometres (Km)
- ★ The standard unit for measuring length is **kilometre**
- ★ The basic nit for measuring length is **metre**

VOLUME

- ★ Volume is the space occupied by an object.
- ★ Volume is measured in **cubic units**

Examples of Cubic units

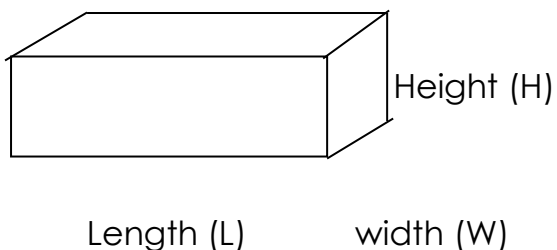
- Cubic metres
- Cubic centimetres (cc)
- Cubic millimetres

Other units of volume apart from cubic units

- Litres
- Millilitres
- Centilitres

FINDING VOLUME OF REGULAR SHAPED OBJECTS

The volume of regular objects like cuboids e.g. boxes blocks bricks etc, can be found after knowing their length, width and height.

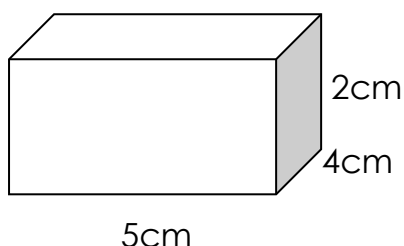


$$\text{Volume} = \text{Length} \times \text{width} \times \text{Height}$$

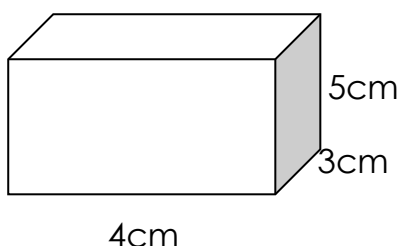
Exercise

1. Find the volume of the figures.

(a)



(b)



2. Find the volume of a brick of length 6cm width 4cm and height 3cm.

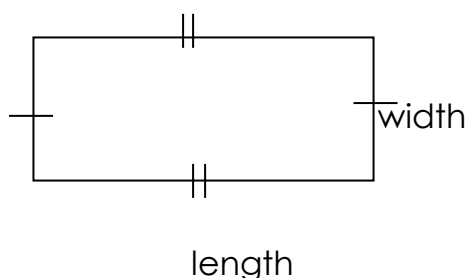
3. Find the area of a cube whose side is 3cm.

CAPACITY

- This is the amount of liquid a container can hold
- Capacity is measured in **litres**

AREA

- Area is the total space occupied by a figure
- Area is measured in **square units**
- In two dimensional figure, the longer side is the length while the shorter side is the width.



IRREGULAR OBJECTS

- These are objects which don't have well defined shape
- These are objects with undefined shape (indefinite shape or improper shape)

Examples of irregular objects

- | | | |
|-----------|----------------|-----------|
| ★ Stones | ★ Broken glass | ★ Padlock |
| ★ Keys | ★ Orange | ★ Pen |
| ★ Needles | ★ Mango | |

Finding volume of irregular objects

- The volume of irregular objects is measured using **measuring by displacement**.

Equipment used when using measuring by displacement method

- ★ Measuring cylinder
To measure the volume of displaced water
- ★ Overflow can (eureka can)

- ★ String (thread)
To lower the irregular object gently into the container
- ★ Water

Containers (instruments) used in measuring by displacement

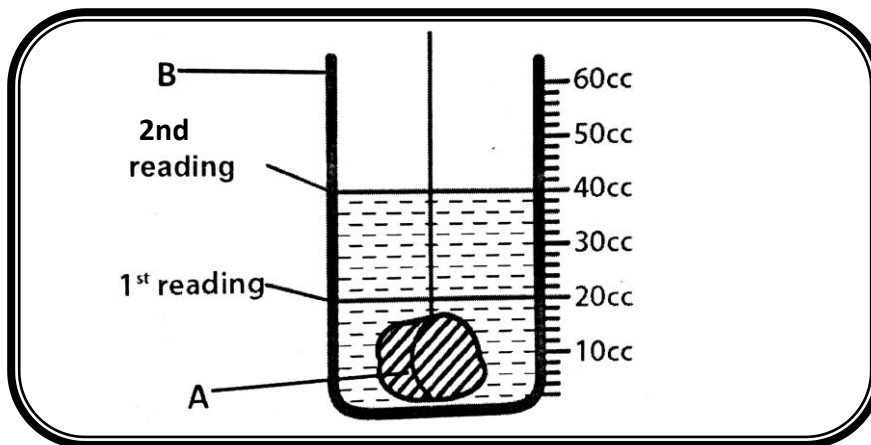
- (a) Measuring cylinder
- (b) Over flow can (Eureka can)

Why is measuring by displacement method called so?

- ★ It involves measuring the volume of displaced water.

Using a measuring cylinder

- ★ Pour water into a measuring cylinder and record its volume.
- ★ Tie a thread/string on an irregular object and lower it into the measuring cylinder.
- ★ Record the new volume of water.
- ★ The difference between volume of water is the volume of the stone.



volume of the stone = 2nd level – 1st level

$$= 40\text{cc} - 20\text{cc}$$

$$= 20\text{cc}$$

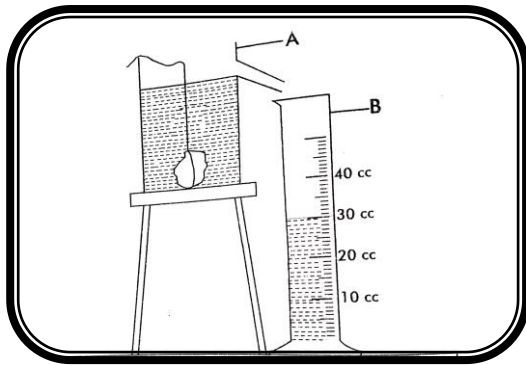
The volume of water is **20cc** because the amount of water displaced is always equal to the volume of the irregular object.

Using an overflow can (eureka can) and measuring cylinder

- ★ Fill the overflow can with water up-to level of the spout
- ★ Lower an irregular object into the overflow can using a thread
- ★ Water will overflow via the spout to the measuring cylinder
- ★ The volume of the irregular object is equal to the volume of water in the measuring cylinder

- ★ The volume of irregular object is the same as the volume of water in measuring cylinder

Find the volume of the stone by reading the level of water in the cylinder.



A – Overflow can/Eureka can

B – Measuring cylinder

The volume of the irregular object is **30cc**.

Weight

- ★ Weight is the force of gravity acting on an object.
- ★ Weight is measured in Newtons (N).
- ★ Weight is not constant (it changes).
- ★ Weight is measured using an instrument called **spring balance**.

Factors that determine weight of an object.

- ★ Size of the object
- ★ Nature of the object
- ★ Upthrust force (buoyancy)
- ★ Force of gravity (gravitational pull)

Note:

Objects weigh less on the moon than on earth **because** the moon's gravity is less than earth's gravity

When lifting up objects, we are overcoming the **force of gravity**.

It is difficult to push a wheel barrow uphill **because** we are opposing the force of gravity.

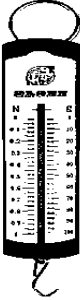
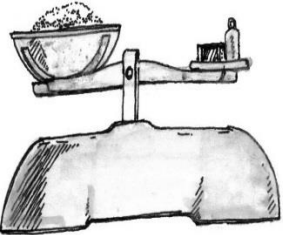
It is easier to push a wheel barrow downhill **because** we are helped by the force of gravity.

MASS

- ★ This is the amount of matter in an object.
- ★ Mass is constant **because** it doesn't depend on the force of gravity.
- ★ The standard unit for mass is **kilogram (Kg)**.
- ★ The basic unit for mass is **gram (g)**.

Examples of machines used for measuring mass

- ★ Beam balance
- ★ Set of scales
- ★ Scale balance
- ★ Lever balance

(a) Beam balance	(b) Spring balance	(c) set of scales	(d) scale balance
			

DIFFERENCES BETWEEN MASS AND WEIGHT

MASS	WEIGHT
Mass is the amount of matter in an object	Weight is the force of gravity acting on an object
Mass is constant	Weight changes
Mass is measured in Kilograms or grams	Weight is measured in Newtons

DENSITY

- ★ Density is the mass of an object per unit volume.
- ★ Density (D) = $\frac{\text{Mass}(M)}{\text{Volume}(V)}$

$$D = \frac{M}{V}$$

Density is measured in kilograms per cubic metre (kg/cm³) or grams per cubic centimetre (g/cm³)

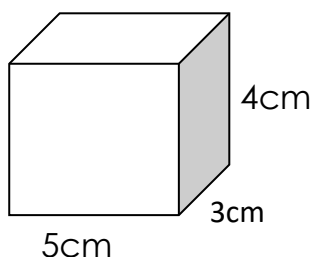
IMPORTANCE OF DENSITY

- ★ It enables us to identify pure and impure substances
- ★ It enables us to find mass of an object
- ★ It enables engineers to use strong metals that are not heavy to make bodies of aeroplanes

Finding Density

1. Find the density of an object of mass 150g and volume 3cc.
2. If the mass of the cuboid is 48g.

Find its density



3. Calculate the volume of a stone of mass 48g and density 6g/cc

DENSITIES OF SOME SUBSTANCES

Substance	Density (g/cc)
★ Gold	★ 19.3
★ Mercury	★ 13.6
★ Lead	★ 11.3
★ Silver	★ 10.5
★ Copper	★ 8.9
★ Brass	★ 8.3
★ Iron	★ 7.8
★ Tin	★ 7.3
★ Aluminium	★ 2.7
★ Glass	★ 2.7
★ Rubber	★ 1.3
★ Salt solution	★ 1.2
★ Water at 4c ⁰	★ 1.0
★ Ice	★ 0.92
★ Methylated spirit	★ 0.83
★ Kerosene	★ 0.8
★ Alcohol	★ 0.8
★ Petrol	★ 0.7
★ Cork	★ 0.25
★ Air	★ 0.0012
★ hydrogen	

The density of pure water is 1 g/cc.

Materials having density of less than 1 g/cc will float on water.

Materials having density of more than 1 g/cc will sink on water.

- ★ Aluminum is used to make bodies of aeroplanes because it has low density (Aluminum is light).
- ★ Objects that are less dense than water will always float on water.
- ★ A substance which is denser than water always sinks into water e.g. iron, brass, nails, stones, sand.

Hydrometer

This is an instrument used for measuring the density of different liquids e.g. milk, beer, wine, acids in batteries.

- ★ A special hydrometer used to measure density of milk is called **lactometer**.

Behaviour of objects when put in water

- ★ Floating
- ★ Sinking

Floating

Floating is when an object remains on top of water or any liquid.

Floating objects

These are objects that remain on the surface of water or any liquid.

- ★ Objects float on water because they are less dense than water.
- ★ The density of water is 1g/cc and therefore, objects whose density is less than 1g/cc float on water.

Liquids that float on water

- | | |
|------------|-------------------|
| ★ Kerosene | ★ Cooking oil |
| ★ Petrol | ★ Lubricating oil |
| ★ Diesel | |

Other things that float on water

- | | |
|---------------|------------|
| ★ Cork | ★ Dry wood |
| ★ Plastic cup | ★ Sponge |
| ★ Ice | ★ Leaf |
| ★ Boats | ★ Feathers |

Objects weigh less in water **due to upthrust force(buoyancy)**.

Sinking

Sinking is when an object goes to the bottom of water.

Sinking objects

These are objects that go to the bottom of water.

- ★ Objects sink in water because they are denser than that water.
- ★ Objects whose density is more than 1g/cc sink in water.

Examples of sinking objects

- | | | |
|-------------|-------------|-----------|
| ★ Stone | ★ Steel bar | ★ Mercury |
| ★ Sand | ★ Glass | |
| ★ Metal rod | ★ Iron nail | |

Note:

A sinking object displaces water equal to its volume while a floating object displaces water equal to its weight.

MISCIBLE AND IMMISCIBLE LIQUIDS

MISCIBLE LIQUIDS

- ★ These are liquids which that can mix easily.
- ★ Miscible liquids form a uniform mixture.
- ★ They have the same density but different boiling points.

Examples of miscible liquids

- | | |
|--|------------------------|
| ★ Water and alcohol | ★ Kerosene and alcohol |
| ★ Miscible liquids are separated by fractional distillation | |

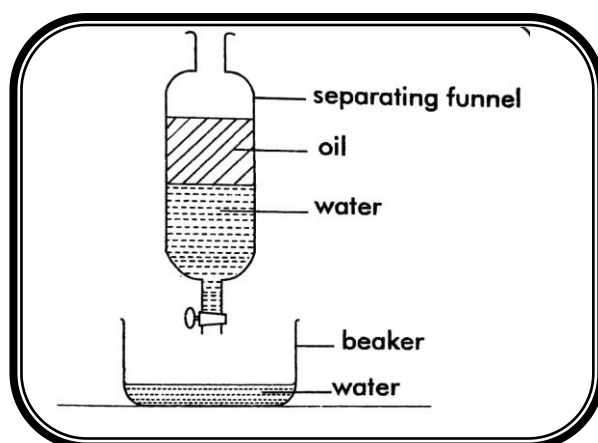
IMMISCIBLE LIQUIDS

- ★ These are liquids that cannot mix
- ★ They cannot form a uniform mixture.

Examples of immiscible liquids

- | | |
|--------------------|---------------------|
| ★ Water and petrol | ★ Mercury and water |
| ★ Oil and water | |

Immiscible liquids are separated using a **separating funnel/floatation method**.



TOPIC THREE: IMMUNIZATION

IMMUNIZATION

- This is the introduction of vaccines into the human body to boost immunity.

Types of immunization

- ★ Primary immunization
- ★ Mass immunization
- ★ Booster immunization

Primary immunization

- This is done to all children below the age of 5 years.

Mass immunization

- This is done to all people in case there is a disease outbreak in an area.

Booster immunization

- This is done to maintain the required amount of antibodies in the human body.

IMPORTANCE OF IMMUNIZATION

- ★ It prevents immunisable diseases.
- ★ It boosts immunity.
- ★ It reduces infant mortality rate.

IMMUNITY

- This is the ability of the body to fight diseases.

Types of immunity

- ★ Natural immunity
- ★ Artificial immunity

NATURAL IMMUNITY

- This is the type of immunity that does not involve use of vaccines.

Ways of acquiring natural immunity

- ★ Through breastfeeding.
- ★ After recovering from sickness (illness).
- ★ Having a balanced diet.
- ★ From the mother to child through the placenta.

Why is it dangerous to acquire immunity after recovering from illness?

- It may lead to body weakness hence causing death.

ARTIFICIAL IMMUNITY

- This is the type of immunity that involves the use of vaccines.

How does the body acquire artificial immunity?

- By immunization.
This is the best way of acquiring immunity.

VACCINES

- These are medical drugs used for immunization.

IMPORTANCE OF VACCINES

Vaccines boost body immunity.

How do vaccines boost immunity?

- They stimulate the production of antibodies. (They enable the body to produce antibodies)

What are antibodies?

- These are substances produced by white blood cells in the body to fight against diseases.

Methods of administering vaccines (methods of immunization)

- ★ Oral method
- ★ Injection method

TYPES OF VACCINES

- ★ Killed vaccines
- ★ Attenuated vaccines
- ★ Toxoid vaccines

ATTENUATED VACCINES

- ★ These are vaccines made from wakened live germs

Examples of live-attenuated vaccines

- ★ Yellow fever vaccine
- ★ Rotavirus vaccine
- ★ Chicken pox vaccine

KILLED (INACTIVATED) VACCINES

- ★ These are vaccines made from killed germs.

Examples of killed (inactivated) vaccines

- ★ IPV (inactivated polio vaccine)
- ★ Rabies vaccine

TOXOID VACCINES

- ★ These are vaccines made toxins produced by germs.

Examples of toxoid vaccines

- ★ TT vaccine (Tetanus toxoid vaccine)

EXAMPLES OF VACCINES

Vaccine	Method of administration	Immunisation site	Disease
BCG vaccine	Injection method	Right upper arm	Tuberculosis (TB)

Vaccine	Method of administration	Immunisation site	Disease
DPT (DTaP) vaccine	Injection method	Left upper thigh	Diphtheria, Pertussis, Tetanus
Measles vaccine	Injection method	Left upper arm	Measles
OPV	Oral method	Mouth	Poliomyelitis (polio)
PCV	Injection method	Right outer thigh	Pneumonia
Rotavirus Vaccine	Oral method	Mouth	Diarrhoea
HPV vaccine	Injection method	Left upper arm	Cervical cancer
Yellow fever vaccine	Injection method	Right upper arm	
Hep B vaccine	Injection method	Left upper thigh	Hepatitis B
hib vaccine	Injection method	Left upper thigh	Haemophilus influenza type B

Why are babies given BCG and polio vaccines at birth?

- Babies are born without immunity against Tuberculosis (TB) and polio.

Why is measles vaccine given at the age of 9 months?

- Babies are born with immunity against measles which lasts for 9 months.

Why was DPT replaced with DTaP?

- DTaP has fewer side effects than DPT.

Write the following vaccines in full

- **OPV:** Oral Polio vaccine
- **IPV:** Inactivated Polio vaccine
- **BCG:** Bacille Calmette-Guerin (Bacillus of Calmette and Guerin)
- **DPT:** Diphtheria, Pertussis and Tetanus
- **DTaP:** Diphtheria, Tetanus and Pertussis (whooping cough)
- **HPV:** human papillomavirus vaccine
- **Hib:** Haemophilus influenza type B

Why DPT (DTaP) is called a triple vaccine?

- ★ It protects the body against three immunisable diseases.

Why is DPT - Hep B + Hib vaccine called a pentavalent vaccine?

- It protects the body against five immunisable diseases.

Why is polio vaccine given very many times?

- Polio virus is very resistant to vaccines.

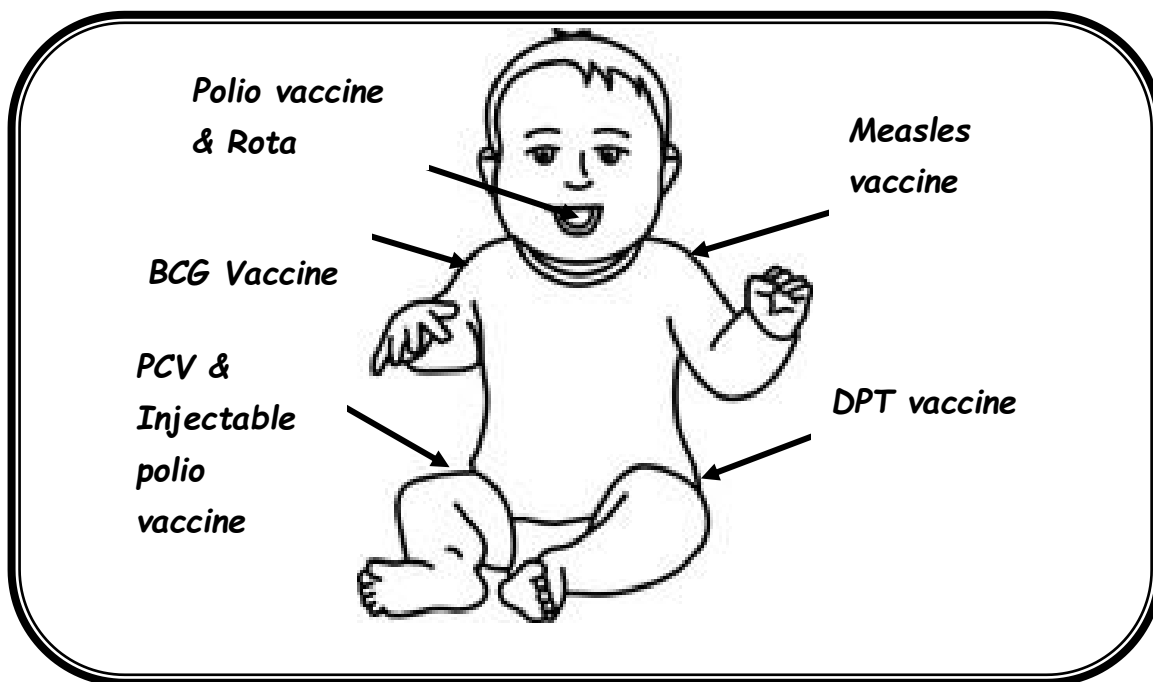
Why is Polio vaccine always given orally?

- It is very effective when given orally than injection.

Immunisation site

- ★ This is the part of the human body where a vaccine is administered.

Immunisation sites on human body



Why are vaccines injected at different parts of the body?

- For easy identification of the disease immunized.

IMMUNISABLE DISEASES

- These are diseases that can be prevented by immunization.

Childhood immunisable diseases

- These are immunisable diseases common in children below the age of 5 years

Examples of childhood immunisable diseases

- | | |
|------------------|---------------------------|
| ★ Measles | ★ Hepatitis B |
| ★ Tuberculosis | ★ Haemophilus influenza B |
| ★ Polio | ★ Diphtheria |
| ★ whooping cough | ★ Tetanus |

SUMMARY TABLE

Disease	Vaccine	How it is administered	Doses	At what age
Tuberculosis	BCG	Injection on upper right arm	1	At birth
Poliomyelitis	Polio vaccine (OPV)	Drops in the mouth	4	At birth 6 weeks 10 weeks 14 weeks
Diphtheria Whooping cough Tetanus Hepatitis – B Influenza – B Haemophilus	DPT Hep – B+ Hib	Injection on the left thigh	3	6 weeks 10 weeks 14 weeks
Measles	Measles vaccine	Injection on the left upper arm	1	9 months (36 weeks)

MEASLES (RUBELLA)

- ★ It is caused by a virus.
- ★ It is an air borne disease.
- ★ It affects the skin and lungs.

How does measles spreads?

- ★ Through inhaling contaminated air.
- ★ Through body contact with sick person.
- ★ Through sharing dirty clothes with sick person.

Signs of measles

- ★ runny nose
- ★ skin rash
- ★ dry cough
- ★ Watery eyes (red eyes)

Symptoms of measles

- ★ Fever
- ★ Sore throat
- ★ Loss of appetite
- ★ Body weakness

Prevention of measles

- Isolate and treat infected ones
- Immunize children against measles.

POLIOMYELITIS (POLIO)

- ★ It is an immunisable waterborne disease.
- ★ It is caused by a virus.
- ★ It affects the skeleton (bones) especially the limbs and the muscles.

Name the germ (virus) that causes polio

- Poliovirus

Which vector spreads poliomyelitis?

- ★ Cockroach

How does poliomyelitis spread?

- Through drinking contaminated water
- Through eating contaminated food

Sign of poliomyelitis

- ★ Paralysis of the limb
- ★ Stiffness of the neck
- ★ Stiffness of the back

Symptoms of poliomyelitis

- ★ Muscle weakness
- ★ Headache
- ★ Vomiting
- ★ Fever
- ★ Neck pain
- ★ Back pain

Effect of poliomyelitis to an individual

- It leads to lameness

Ways of preventing and controlling poliomyelitis

- ★ Immunisation using polio vaccines (IPV & OPV).
- ★ Drinking boiled water.
- ★ Proper use of latrines (proper disposal of human wastes).
- ★ Wash hands with clean water and soap before eating food.

How are crutches useful to a polio victim?

- Crutches help the polio victim to move.

TUBERCULOSIS

- ★ It is an immunisable airborne disease (droplet infection).
- ★ It mainly affects the lungs and spine (backbone).
- ★ It affects the respiratory system and skeletal system.
- ★ It is caused by a bacterium.

Name the germ (bacterium) that causes tuberculosis

- Mycobacterium tuberculosis

How does tuberculosis spread?

- ★ Through inhaling contaminated air.
- ★ Through drinking contaminated milk from tubercular (sick) cows.
- ★ Through sharing forks, cups and plates with an infected person.

Signs of tuberculosis

- ★ Chronic cough
- ★ Loss of weight
- ★ Severe sweating at night (severe night sweats)
- ★ Thick sputum with blood

Symptoms of tuberculosis

- General body weakness
- Backache
- Mild fever

Ways of preventing and controlling tuberculosis

- ★ Immunization using BCG vaccine.
- ★ Isolate and treat the infected persons.
- ★ Drink boiled or pasteurized milk.
- ★ Don't share forks, cups and plates with sick person.

How is smoking related to TB?

- ★ Smoking worsens tuberculosis

Name the Sexually Transmitted Disease (STD) always mistaken for Tuberculosis (TB)

- ★ AIDS

Tetanus

- ★ It is caused by bacteria found in the soil
- ★ It affects the muscles
- ★ The bacteria enters the body through fresh cuts or dirty wounds.
- ★ In new born babies, it can enter through the umbilical cord if it is cut with a dirty instrument like a razor blade or knife

Signs and symptoms

- ★ Muscles get stiff
- ★ Spasms when touched
- ★ The baby stops breast feeding

Why is tetanus called LOCK JAW disease?

- ★ It makes the jaws of the baby to become stiff.

Prevention and control

- ★ Immunise children against tetanus (using DPT vaccine).
- ★ Keep the wounds clean all the time.
- ★ Immunise expectant mothers (pregnant women) with TT vaccine.
To protect them and the unborn babies from tetanus infection.

To enable the unborn baby get immunity against tetanus.

- ★ Immunise teenage girls with TT vaccine.
To prevent them from getting tetanus (during menstruation).

Groups of people who are given TT vaccine

- ★ Teenage girls
- ★ Pregnant women
- ★ Accident victims

Why is DPT vaccine given to babies at 6 weeks?

- ❖ Babies are born with maternal immunity that lasts six weeks.

Whooping cough (pertussis)

- ★ It is caused by bacteria.
- ★ It is an airborne disease (droplet infection).
- ★ It spreads through inhaling contaminated air.
- ★ It affects the lungs.

Signs and symptoms

- ★ Severe cough (coughing spells)
- ★ Gasps for breath
- ★ Runny nose

Prevention and control

- ★ Immunise babies using DPT vaccine.

Diphtheria

- ★ It is caused by bacteria.
- ★ It affects the nose and throat.
- ★ It spreads through inhaling contaminated air.

Signs and symptoms

- ★ Sore throat
- ★ swollen neck

Hemophilus influenza type B

This disease is caused by a virus.

It is prevented using Hib vaccine.

Hepatitis B

It is caused by a virus.

It affects the liver.

How does hepatitis B spread?

Through playing unprotected sex with infected person.

Through body contact with infected body fluids.

Signs of hepatitis B

Dark urine

Vomiting

Jaundice (yellow eyes and soles of feet)

Prevention of hepatitis B

Immunise with Hep B vaccine.

Avoid playing unprotected sex with infected person.

Avoid sharing needles with infected person.

Other immunisable

- ★ cholera
- ★ meningitis
- ★ Yellow fever
- ★ Diarrhoea
- ★ Pneumonia
- ★ Cervical cancer (affects the female reproductive system)
- ★ German measles (rubella)
- ★ rabies
- ★ Small pox

Effects of immunisable diseases

- ★ Increased death rate of infants
- ★ Lameness
- ★ Mental disabilities

Roles of individuals, families and communities in immunisation

Parents

- ★ Taking their children for immunisation.
- ★ Participating in community immunisation activities and programme.
- ★ Assisting schools to identify children who are not immunized.

Children

- ★ Taking their siblings (young brothers and sisters) for immunization.
- ★ Informing parents about the day for immunization.
- ★ Reciting songs and plays on immunization.
- ★ Helping to identify children who have not been immunized in the community.

Government

- ★ Announcing immunisation centres.
- ★ Announcing outbreak of immunisable disease.
- ★ Announcing immunisation days.
- ★ Educating people about immunization.
- ★ Funding immunisation programmes.
- ★ Distributing vaccines to health centres.

Community leaders

- ★ Inviting health workers to educate members about immunisation
- ★ Encouraging members to organize immunisation centres
- ★ Encouraging members to take their children for immunisation
- ★ Reporting immunisable disease outbreak

Child to child programme

This is the health programme where older children help their siblings to solve health problems.

Child health card (Immunisation card)

This is a document that contains child's health information given to every child receiving immunisation.

Information on the child health card

- ★ Child's birth weight
- ★ Child's name
- ★ Child's date of birth
- ★ Growth graph of a child
- ★ Sex of the child
- ★ Next date for immunisation
- ★ Diseases immunised
- ★ Mother's name
- ★ Father's name
- ★ Parents' occupation
- ★ Place of residence

Importance of child health card

- ★ It helps a parent to monitor the child's growth.
- ★ It helps the parent to know the next date for immunisation.
- ★ It helps the doctor to know diseases immunized.
- ★ It helps the doctor to know which dose is remaining.

How can one identify whether a baby was immunised against TB or not ?

- ❖ By checking the child health card.
- ❖ By checking the injection scar on the right upper arm.

Government and NGOs which help to carry out immunisation activities in Uganda

- ★ WHO
WHO stands for World Health Organisation
- ★ UNICEF
UNICEF stands for United Nations Children's Fund
- ★ UNEPI
UNEPI stands for Uganda National Expanded Programme on Immunization
It is under Ministry of Health

Role of UNEPI

- ★ It organises National Immunisation Days.
 - ★ It provides immunisation equipment.
 - ★ It distributes vaccines to health centres.
 - ★ It educates people about immunization.
 - ★ It monitors the national immunization programme.
- NIDs** stands for National Immunisation Days.

Other organizations that support UNEPI in immunization.

- ★ Churches
- ★ Schools
- ★ Rotary clubs
- ★ Save the Children's Fund (SCF)

Why has the government of Uganda made immunisation free of charge?

- ❖ For all ugandans to be immunised

Reasons why children in some families are not immunised.

- ★ Ignorance of immunisation
- ★ Cultural beliefs in some families
- ★ Limited access to immunisation services

Name the vaccine that prevents Neonatal tetanus

- ❖ Tetanus Toxoid vaccine

Activity

1. What is immunization?
2. Define the following terms
 - a) Immunity
 - b) Antibodies
 - c) Immunisation site
 - d) vaccines
3. Name the two types of immunisation
4. What type of immunity does a baby acquire from the mother through breast feeding ?
5. Give the meaning of artificial immunity.
6. State two ways the body may acquire natural immunity
7. How does the body acquire artificial immunity?
8. State two methods of administering vaccines in the body
9. Name the vaccine given to children to prevent diarrhoea.
10. How is polio vaccine administered?
11. Name vaccine administered to babies at the age of 36 weeks.
12. Name the vaccine that provides immunity against Diphtheria, whooping and Tetanus.
13. Why is DPT(DTaP) vaccine called triple vaccine?
14. State any two importance of a child health card.
15. Write UNEPI in full.
16. Name any two domestic animals that are vaccinated (Immunised)
17. State two reasons why parents take their children for immunization.
18. Why are expectant mothers given Tetanus Toxoid vaccine?
19. State one vaccine that is given as single dose to a baby.
20. Name one vaccine that is administered orally besides polio vaccine
21. Why is polio vaccine given to a child at birth?
22. Why is measles vaccine given after nine months?
23. How can a P.5 pupil participate in immunization campaign?
24. Name one type of vaccine
25. Why do some families fail to take children for immunization?

TOPIC FOUR: DIGESTION

BODY SYSTEMS

- ★ A system is a group of organs that perform the same function.
- ★ An organ is a group of tissues that perform the same function.
- ★ A tissue is a group of cells that perform the same function.
- ★ A cell is the smallest unit of life.

DIGESTIVE SYSTEM

- This is the body system that breaks down food into small soluble particles that can be absorbed into the body.

DIGESTION

- This is a process by which food is broken down into soluble particles that can be absorbed in into the body..

- It begins in the **mouth** and ends in the **ileum**

Importance of food digestion

- ★ It breaks down food for easy absorption of nutrients.

Types of food digestion

- Mechanical digestion
- Chemical digestion

MECHANICAL DIGESTION

- This is the physical breakdown of food into smaller particles.
- It is done by the teeth and stomach walls.

Forms of mechanical digestion

- Chewing of food
- Churning of food

CHEMICAL DIGESTION

- This is the breakdown of food into soluble particles by the help of enzymes.

How does chemical digestion depend on mechanical digestion?

- ★ Mechanical digestion increases the surface area of food for the action of enzymes.

ENZYMES

- These are chemical substances that speed up chemical digestion in the body.

Importance of enzymes

- ★ They speed up chemical reactions/digestion.
- ★ Enzymes act as catalysts in chemical reactions.

Characteristics of enzymes

- ★ They are specific in action.
- ★ They work at specific pH.
- ★ They are destroyed (denatured) by heat.
- ★ They work under narrow temperature range.
- ★ They are needed in minute amounts.

Why are enzymes denatured (destroyed) by heat?

- ★ They are proteins in nature.

Types of enzymes

- ★ Amylase
- ★ Lipase
- ★ Protease

Conditions under which enzymes work

- ★ Alkaline conditions
- ★ Acidic conditions

What is meant by pH?

- This is the degree of acidity or alkalinity.

Examples of enzymes and the medium (pH) in which they work best

Enzyme(s)	Medium of action (pH)
• Pepsin • Rennin	• Acidic medium
• Salivary amylase • Lipase • Trypsin • Peptidase (erepsin) • Sucrase	• Alkaline medium

Alimentary canal

- This is the long muscular tube that runs from the mouth to the anus
- It is up to 10 metres long.

How does food move in the alimentary canal?

- ★ By peristalsis

What is peristalsis?

- ★ This is the wave-like movement of food through the alimentary canal

Parts of the human digestive system

- ★ gullet (oesophagus)
- ★ liver
- ★ stomach
- ★ pancreas
- ★ gall bladder
- ★ bile duct
- ★ Duodenum
- ★ ileum
- ★ large intestine (colon)
- ★ appendix
- ★ rectum
- ★ Anus

Parts of the alimentary canal

- ★ Gullet (oesophagus)
- ★ stomach
- ★ duodenum
- ★ colon (large intestine)
- ★ ileum
- ★ rectum
- ★ anus

Parts of the digestive system besides the alimentary canal

- ★ Liver
- ★ Pancreas
- ★ Gall bladder

Name the three major glands of the digestive system

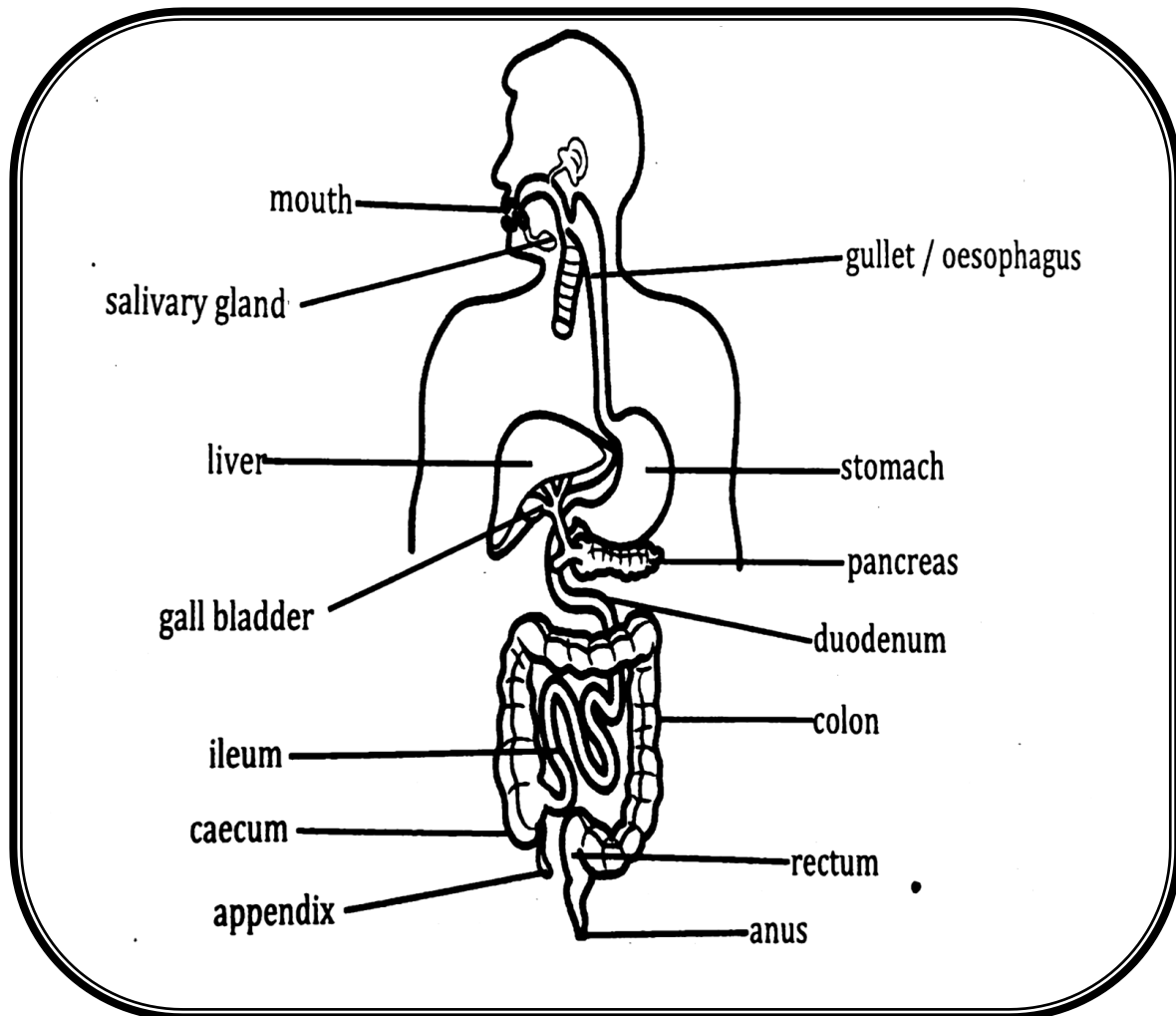
- ★ Salivary glands
- ★ Liver
- ★ Pancreas

Activity

1. How does digestion differ from digestive system?
2. Where does digestion of food
 - i) Begin?
 - ii) End?
3. Name the two types of digestion
4. Mention any one example of mechanical digestion

5. State the role of enzymes in chemical digestion.
6. Name any two enzymes that work under acidic pH
7. How does eating hot food affect the chemical digestion?

THE STRUCTURE OF A HUMAN DIGESTIVE SYSTEM



DIGESTION OF FOOD IN THE MOUTH

- ★ The teeth break down food.
- ★ Chewing increases the surface area of food for the action of enzymes.

SALIVA (SALIVARY JUICE)

- ★ This is the digestive juice found in mouth.
- ★ It is produced by **salivary glands**.
- ★ It contains an enzyme called **salivary amylase (ptyalin)**.
- ★ Salivary amylase works in alkaline conditions.
- ★ Salivary amylase works on carbohydrates.
- ★ Salivary amylase changes cooked starch to maltose.
- ★ Digestion of carbohydrates begins in the mouth!

Importance of saliva

- ★ It moistens/softens food.
- ★ It lubricates food for easy swallowing.
- ★ It cools hot food.
- ★ It contains salivary amylase that works on cooked starch.

Why does the action of salivary amylase (ptyalin) stop when food reaches the stomach?

- Salivary amylase cannot work under acidic conditions in the stomach.
- There are no alkaline conditions for salivary amylase to work.

IMPORTANCE OF THE TONGUE

- It rolls food into bolus (small ball).
- It mixes food with saliva.
- It pushes food to gullet.

What prevents food from entering the wind pipe (trachea) during swallowing?

- Epiglottis

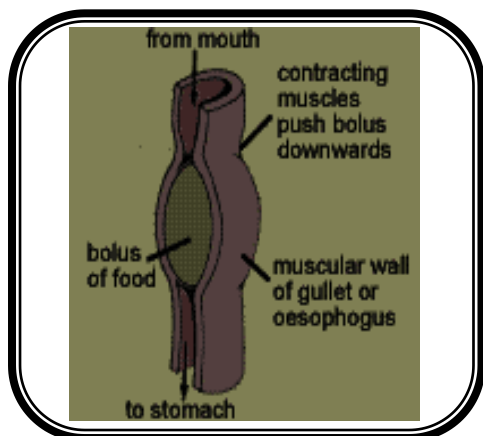
FOOD IN THE GULLET (OESOPHAGUS)

- Gullet is the tube that directs food from the mouth to the stomach.
- Food moves from the gullet to the stomach by **peristalsis**.

Importance of the gullet

- ★ It passes (directs) food to the stomach

A diagram showing peristalsis in the gullet



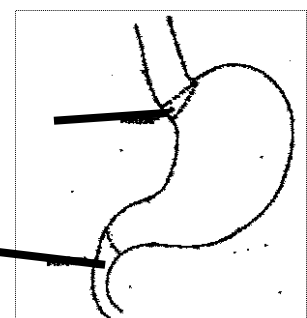
FOOD DIGESTION IN THE STOMACH

- ★ Stomach is a muscular bag that stores food for some time.
- ★ Food stays in the stomach for about 3 hours.
- ★ **Cardiac muscles** allow food into the stomach.
- ★ The stomach walls mixes food with digestive juices to form a chyme.
- ★ **Chyme** is the mixture of partly digested food and digestive juices in the stomach.
- ★ **Pyloric sphincter muscles** allow chyme into the duodenum.
- ★ The stomach has acidic conditions.
- ★ Chemical digestion of proteins begins in the stomach.

Simple diagram to illustrate (The stomach)

Cardiac sphincter muscle

Pyloric sphincter muscle



Importance of stomach

- ★ It stores eaten food for some time.
- ★ It churns food.
- ★ It is where digestion of proteins begins.

Substances that are absorbed in the stomach

- ★ Simple sugars
- ★ Common salt
- ★ Medicines
- ★ Alcohol

Things produced by the stomach walls

- ★ Gastric juice
- ★ Hydrochloric acid
- ★ Mucus

1. GASTRIC JUICE

- This is the digestive juice found in the stomach

Name the two digestive enzymes in gastric juice

- Pepsin
- Rennin

Pepsin and Rennin work in acidic conditions

IMPORTANCE OF PEPSIN AND RENNIN

Pepsin

- It breaks down proteins into peptides.

Rennin (chymosin)

- It clots milk proteins in the stomach of infants into casein.
- It curdles (coagulates) milk in the stomach of infants.

How important is clotting of milk proteins in babies?

- ★ It allows proper digestion of proteins.
- ★ It helps to separate proteins from fats.
- ★ It extends the period at which milk is retained in the stomach.

2. HYDROCHLORIC ACID

- ★ It provides acidic conditions for pepsin enzymes to digest proteins.
- ★ It activates pepsin enzymes.
- ★ It stops action of ptyalin enzymes on cooked starch.
- ★ It kills germs taken in with food.

3. MUCUS

- ★ It protects stomach walls from hydrochloric acid and enzymes.

THE SMALL INTESTINES (SMALL BOWEL)

- ★ It lies between the stomach and the large intestines.

Parts of the small intestines

- ★ **Duodenum:** it is the upper part
- ★ **Jejunum:** it is the middle part
- ★ **Ileum:** it is the lower part

FOOD IN THE DUODENUM

- ★ The duodenum is the first part of the small intestines.
- ★ It is U-shaped and about 25cm long.
- ★ It is where digestion of fats begins.
- ★ No food absorption occurs in the duodenum.
- ★ It contains pancreatic juice and bile juice (bile).
- ★ It receives bile juice through the bile duct.
- ★ It receives pancreatic juice through the pancreatic duct.

Digestive juices found in the duodenum

- ★ Pancreatic juice
- ★ Bile juice (bile)

PANCREAS

- ★ This is a digestive organ/gland below the stomach.

Functions of pancreas

- ★ It produces pancreatic enzymes.
- ★ It produces insulin hormone that regulates sugar in blood.

Enzymes found in pancreatic juice (duodenum)

- ★ Pancreatic amylase - carbohydrates
- ★ Lipase - fats and oils
- ★ Trypsin - proteins

Enzymes in pancreatic juice work under alkaline condition

Pancreatic amylase

- It changes starch to maltose.

Lipase

- It changes fats and oils (lipids) to fatty acids and glycerol.

Trypsin

- It changes peptides to amino acids.

Importance of the pancreas

- ★ It produces pancreatic juice.

BILE JUICE (BILE)

- ★ This is a digestive juice which has no enzymes.
- ★ It is produced by the liver and stored in the gall bladder.
- ★ It does not contain any enzyme.

Functions of bile during food digestion

- ★ To breakdown (emulsify) fats in the duodenum.
- ★ To neutralize acidic chyme from the stomach.
- ★ To provide alkaline conditions in duodenum.

GALL BLADDER

- This is a small pouch that sits under the liver.
- The gall bladder is full of bile before a meal and it empty after a meal.

IMPORTANCE OF GALL BLADDER

- ★ To store bile juice (bile).
- ★ To control the flow of bile juice into the duodenum.

FOOD IN THE ILEUM

- ★ The ileum is the lowest part of the small intestines.
- ★ The ileum is about 6 metres long.
- ★ Food digestion ends in the ileum.
- ★ Food absorption occurs in the ileum.
- ★ The ileum has finger-like structures/projections called **villi** that absorb digested food
- ★ The walls of the small intestines produce (secrete) a digestive juice called **intestinal juice (succus entericus)**.

Digestive enzymes found in intestinal juice (succus entericus)

- | | |
|-----------|-----------------------|
| ★ Maltase | ★ Lipase |
| ★ Lactase | ★ Peptidase (erepsin) |
| ★ Sucrose | |

Processes that occur in the ileum

- ★ Food digestion
- ★ Food absorption (it is the main process)

Main material absorbed in the ileum

- ★ Digested food (food)

Food absorption

- ★ This is the process by which digested food is taken into the blood stream.

ADAPTATIONS OF THE ILEUM TO ITS FUNCTIONS

- ★ It is long.
To increase surface area for food absorption.
- ★ It has many villi
To increase surface area for food absorption.
- ★ It has thin walls (thin epithelium).
For easy diffusion of digested food.

- ★ It has a dense network of blood capillaries.
For easy diffusion of digested food.
- ★ It is coiled.
To increase surface area for food absorption.
- ★ It has narrow lumen.
To slow food movement that allows proper absorption.

Why does digestion of food end in the ileum?

- ★ It has many enzymes to complete food digestion.
- ★ It has villi to absorb digested food.

HEPATIC PORTAL VEIN

- It carries blood with digested food from the ileum to the liver.

Why does blood with digested food (blood from the ileum) go to the liver?

- For the blood to be detoxified (for the liver to remove toxic substances).
- For the liver to store excess nutrients.

THE LIVER

- This is a reddish brown organ in the abdominal cavity below the diaphragm.
- It is the largest internal organ in the human body.

FUNCTIONS OF THE LIVER

- ★ It produces bile juice (bile) which breaks down fats and oils.
- ★ It detoxicates blood (removes toxic substances from blood).
- ★ It helps in deamination (converts excess amino acids into urea).
- ★ It regulates blood sugar level storing excess glucose as glycogen.
- ★ It produces body heat.
- ★ It stores some **vitamins** (A, D, E and K) and **mineral salts** (iron and potassium).

Why does a dead body (corpse) feel cold?

- ★ The liver that produces heat has stopped working.

Why is a liver called a storage organ?

- It stores some vitamins and mineral salts.

Name three toxic (harmful) substances removed from blood by the liver

- ★ Alcohol
- ★ Urea
- ★ Expired drugs

FOOD VALUES AND THEIR END PRODUCTS

Food value	End products
▪ Carbohydrates	Glucose
▪ Proteins	Amino acids
▪ Fats and oils	Fatty acids and glycerol

Uses of digested food by our bodies

- ★ Glucose (from carbohydrates) provides energy to the body.
- ★ Amino acids (from proteins) build/helps in proper growth of the body.
- ★ Amino acids repair worn out body tissues.
- ★ Fatty acids and glycerol (from lipids) provide heat and energy to the body.

THE LARGE INTESTINES (LARGE BOWEL)

- ★ It is about 1.5 metres long.

Parts of the large intestines

- ★ Caecum
- ★ Colon
- ★ Rectum
- ★ Anus

CAECUM

- ★ It connects the small intestines to the colon.

COLON

- ★ It absorbs water and salts.

Main process that takes place in the colon

- ★ Absorption of water.

Materials absorbed in the large intestines (colon)

- ★ Water
- ★ Mineral salts (Sodium and potassium)
- ★ Vitamin K

Apart from the colon (large intestines), where else does water absorption take place?

- ★ In the ileum

RECTUM

- ★ The rectum stores undigested food before it is passed out.
- ★ It also stores worn out cells from the digestive tract.

ANUS

- ★ It passes out faeces/stool/excrement.

Egestion (defecation)

- ★ This is the elimination (passing out) of faeces from the body.

Materials that make up faeces (components of faeces)

- ★ Roughages (Dietary fibres)
- ★ Water
- ★ Dead bacteria
- ★ Dead cells

Activity

1. Explain the term absorption of food?
2. Where does absorption of food take place?
3. How is the ileum adapted to food absorption?
4. Where does absorption of the following foods take place?
 - i) Alcohol
 - ii) Water
5. Where does digestion of fats take place in the mouth?
6. Write down things that happen to food in the mouth?
7. By what process does food move down the gullet?
8. Identify the two enzymes produced in the stomach.
9. State the major function of the following parts of digestive system.
 - i) Gall bladder
 - ii) Liver
10. Name the enzyme that breaks down carbohydrates into maltose
11. What is the major role of bile produced by the liver in the digestive process?
12. Identify the major function of a spleen.
13. Identify the digestive juice produced by;
 - i) Pancreas
 - ii) stomach

SUMMARY TABLE OF DIGESTIVE PROCESSES

Part	Gland	Digestive juice	Enzymes	Food changes
Mouth	Salivary glands	saliva	Salivary amylase (Ptyalin)	Starch to maltose
Stomach	Gastric	Gastric juice	pepsin	Proteins to peptides

Part	Gland	Digestive juice	Enzymes	Food changes
	glands		Rennin	clots milk proteins in the stomach of infants (curdles/coagulates milk the stomach of infants)
Duodenum	liver	bile	No enzyme	Emulsifies the fats
	pancreas	Pancreatic juice	Trypsin	Proteins to peptides Peptides to amino acids
			Pancreatic amylase	Starch to maltose
			Lipase	Lipids (fats) to fatty acids and glycerol
Ileum		Intestinal juice (succus entericus)	Lactase	Lactose to glucose/galactose
			Maltase	Maltose to glucose
			Sucrase	Sucrose to glucose/fructose
			Lipase	Fats to fatty acids and glycerol
			peptidase	Peptides to amino acids
Colon				Water absorption
Rectum				Storage of faeces

DIGESTIVE DISORDERS (disturbance to alimentary canal and make it fail to function properly)

- ★ Constipation
- ★ Indigestion
- ★ Vomiting
- ★ Intestinal obstruction

CONSTIPATION

- ★ This is a condition where by a person finds it difficult to pass out faeces. The faeces are very hard and dry and do not come out easily.

Causes

- ★ Lack of roughages in one's diet.
- ★ Having irregular meals.
- ★ Drinking too little water.
- ★ Lack of exercise to the body.

INDIGESTION

- ★ This happens when food is not properly digested.
- ★ A person feels stomach pains, heart burn (burning in the chest) and tiredness.

Causes

- ★ No chewing food properly
- ★ Eating so hurriedly
- ★ Too much drinking of alcohol

INTESTINAL OBSTRUCTION

- ★ Caused when intestine twist or fold themselves causing vomiting, thirsty and death if not reported immediately to the doctor.

VOMITING

- ★ This is a sign of very many diseases.
- ★ It disturbs the digestive system and forces the cardiac sphincter to open and the food in the stomach to be ejected through the mouth.
- ★ Seek for medical attention.

DISEASES OF THE DIGESTIVE SYSTEM

- ★ Appendicitis
- ★ Peptic ulcer
- ★ cholera
- ★ Typhoid
- ★ Diarrhoea

- ★ Dysentery
- ★ Hepatitis

Appendicitis

- ★ This is the inflammation of the appendix.
- ★ This may due to stones or other indigestible solids that get trapped in the appendix.
- ★ See the doctor for medical attention.

Peptic ulcers

- ★ These are sores in the stomach wall caused by too much acid.
- ★ The victim experiences a chronic sharp pain in the stomach and frequent heart burn.

Dysentery

- ★ It is caused by a bacterium or Amoebas got from uncooked food or dirty food or water.
- ★ The victim visits the toilet very frequently, passing out watery stools with some blood.

Typhoid

- ★ Caused by bacteria which cause inflammation of the intestine and easily lead to death.

Ways through which food is made dirty

- ★ Handling food with dirty hands.
- ★ House flies landing on food with their hairy bodies.
- ★ Use of dirty containers to keep food.
- ★ Using dirty utensils to handle food.
- ★ Leaving food uncovered.
- ★ Preparing food in dirty places.

Good eating habits

- ★ Washing hands with soap before serving, handling or eating food.
To kill germs in hands.
- ★ Washing hands after visiting a latrine/Toilet.
- ★ Chewing food properly before swallowing .
To prevent constipation.
- ★ Avoid talking while eating food.
To prevent choking.
- ★ Feeding on a balanced diet.
- ★ Wash hands after eating.

- ★ Washing fruits/vegetables before eating them.
To prevent hookworm infestations.
To remove germs.
- ★ Eat well cooked food.
To ease digestion
- ★ Brush teeth every after a meal.
- ★ Do not eat contaminated.
To prevent food poisoning and diarrhoeal diseases.

Bad eating habits

- ★ Eating food with unwashed hands.
- ★ Opening the mouth while eating food.
- ★ Eating hurriedly.
- ★ Talking while eating.
- ★ Opening bottle tops using the teeth.

How to improve on the working condition of digestive system

- ★ Eat and serve food with clean hands.
- ★ Have regular physical exercises.
- ★ Avoid eating stale or rotten food.
- ★ Having a balanced diet.
- ★ Avoid drinking too much alcohol.
- ★ Including enough roughages in the diet.
- ★ Have regular meals.
- ★ Always eat the right quantity of food.
- ★ Avoid eating while talking.

Activity

1. Write down three diseases and three disorders of digestive system.
 - a. diseases
 - b. disorders
2. Suggest any two ways through which food we eat gets contaminated.
3. Write down at least two examples of good eating habits.
4. Why should food be covered immediately after being cooked ?
5. How are house flies adapted to spreading of diseases?